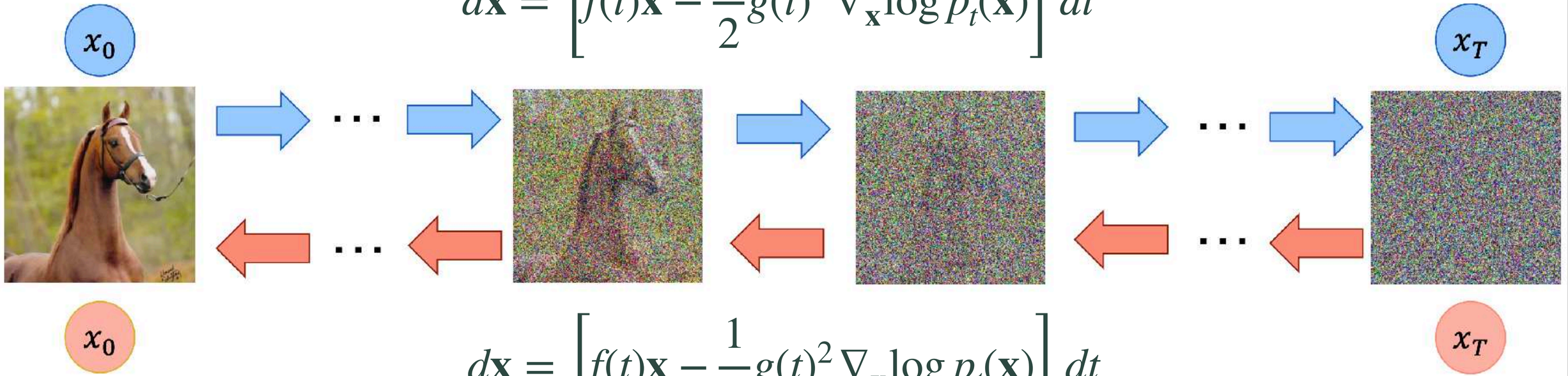

ODE-Free Few-Step Models!
COMPLEX ODE SOLVER
Taking forever!
So... many... steps...
Woohoo!
FAST RESULTS
We're done already!
Nice and quick!

Probability Flow ODE

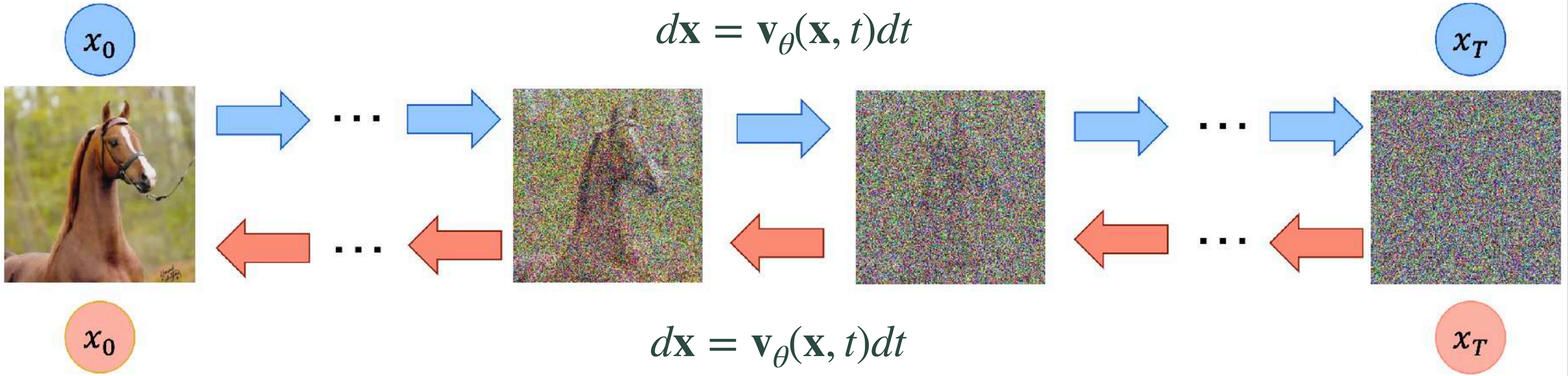
$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$



$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

Y. Song, et al. Score-based generative modeling through stochastic differential equations. 2021

Flow Matching



Numerical solvers offer 15-20 NFE sampling. What about 1-4 steps?

Y. Lipman, et al. Flow Matching for Generative Modeling. 2022

Few-step Models

Flow Map Models (ODE Integrators)

Knowledge distillation

Consistency Models

Consistency Trajectory Models

Shortcut Models

Mean Flow

ODE-free Few-step Models

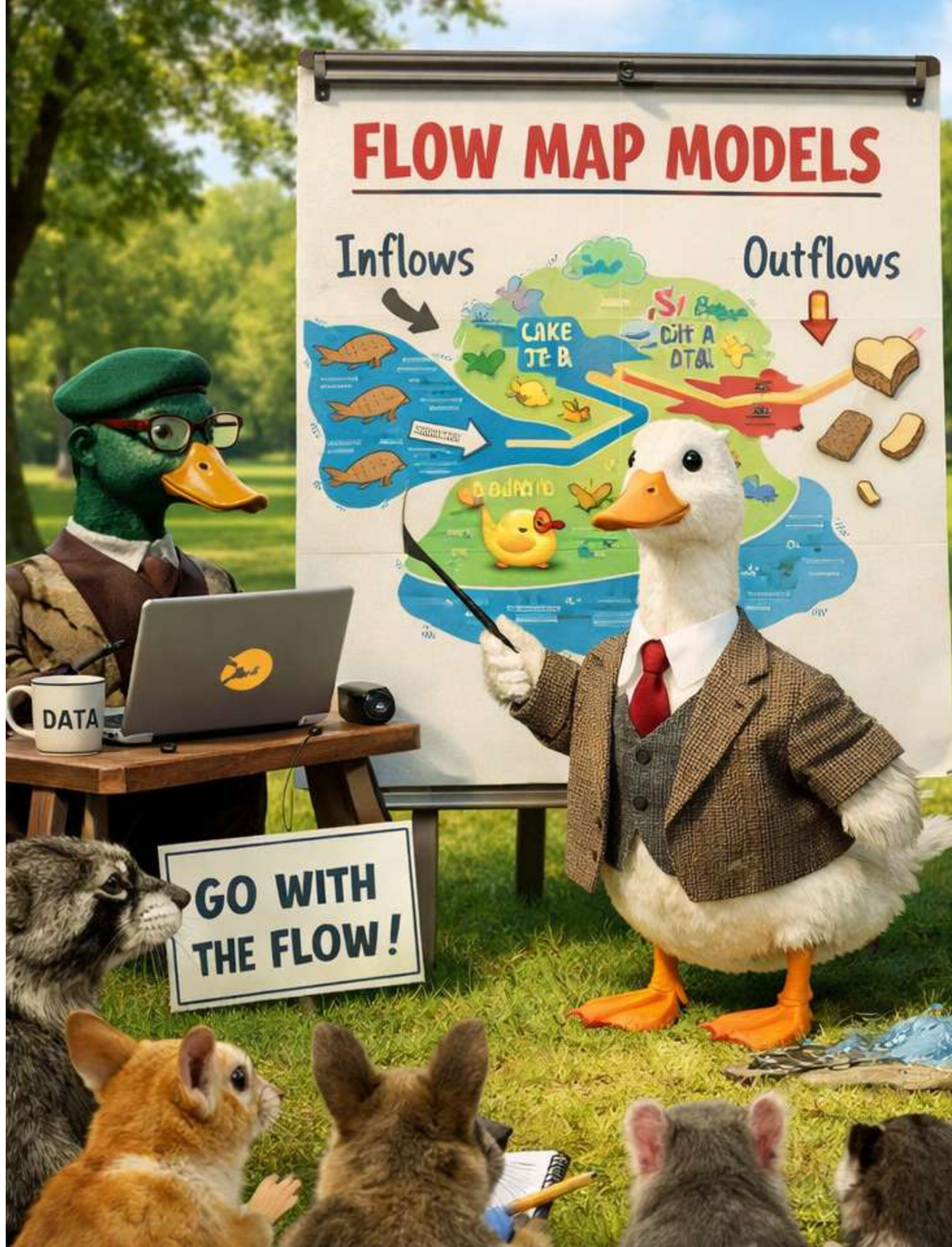
Distribution Matching Distillation

Adversarial Diffusion Distillation

MMD Distillation

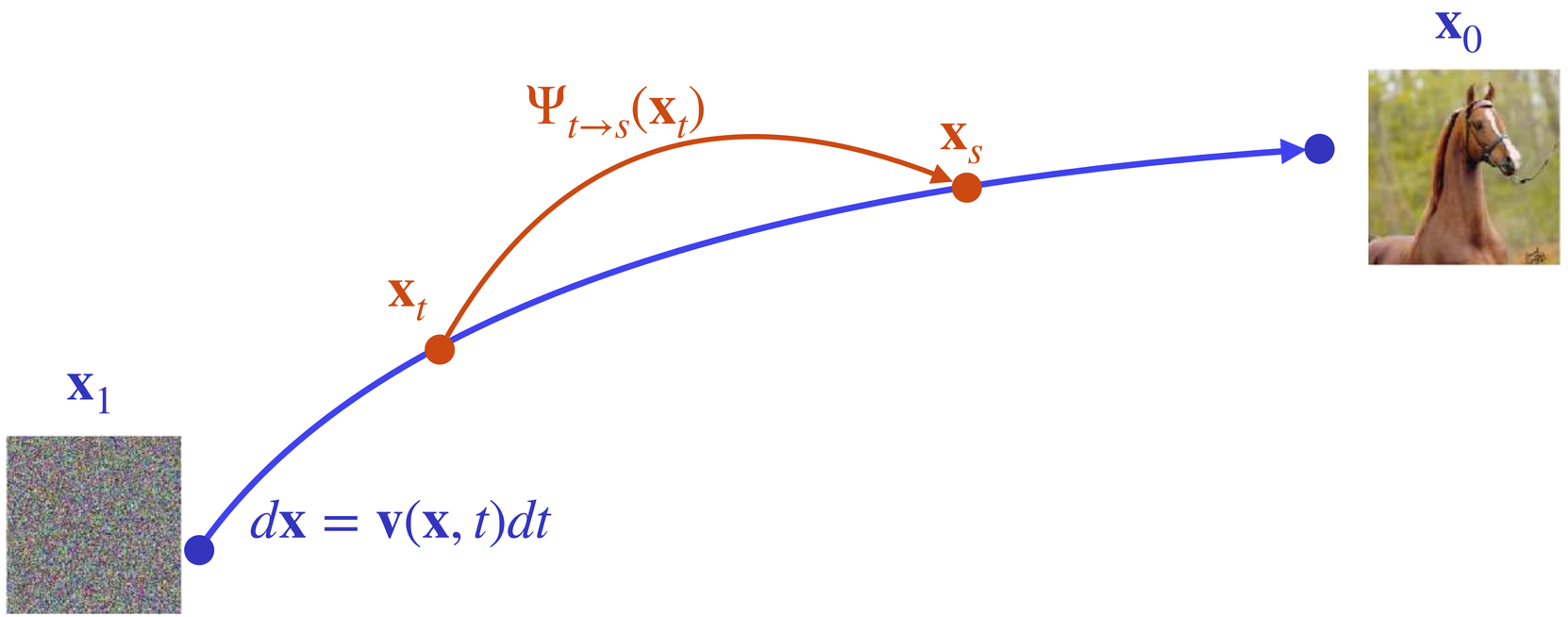
Drifting Models

Recap of Flow Map Models



Flow-Map Formulation

Flow Map: $\Psi_{t\to s}(\mathbf{x}_t) = \mathbf{x}_t + \int_t^s \mathbf{v}(\mathbf{x}, u)\, du$ $\rightarrow$ Oracle integrator for $d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$



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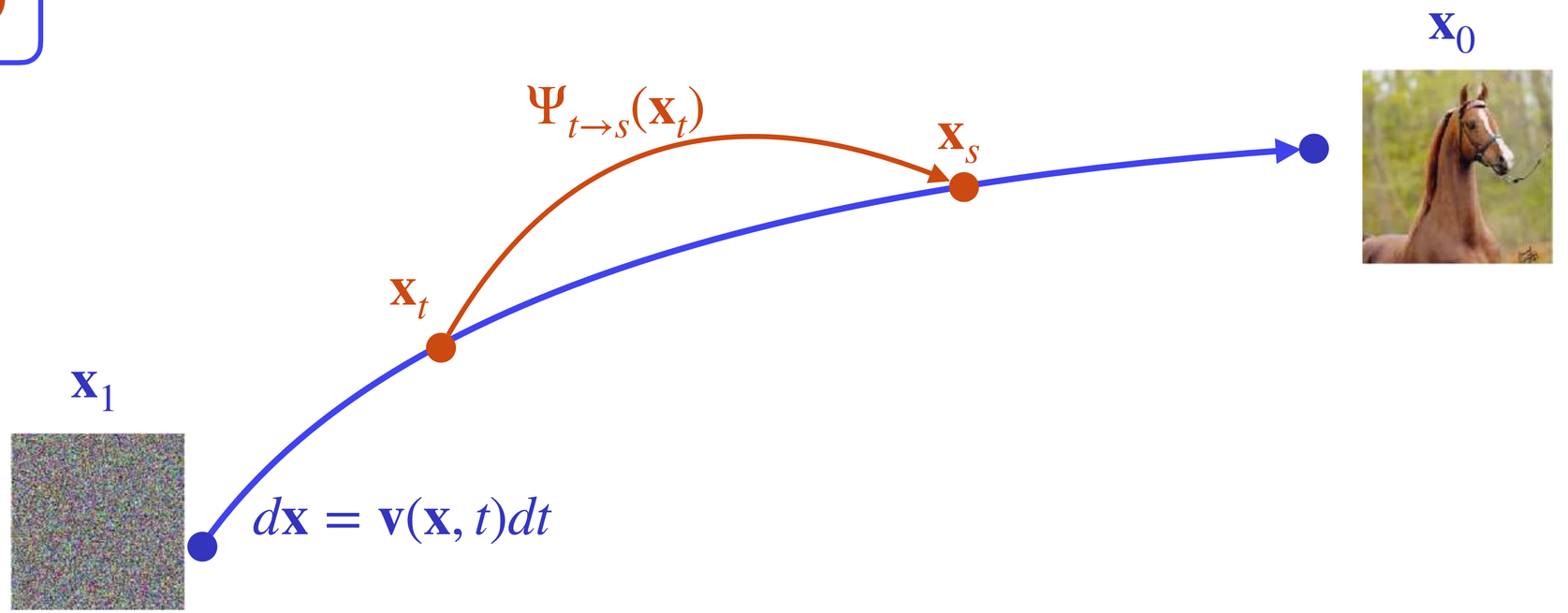
Goal: train $F_\theta(\mathbf{x}_t, t, s) \approx \Psi_{t\to s}(\mathbf{x}_t)$



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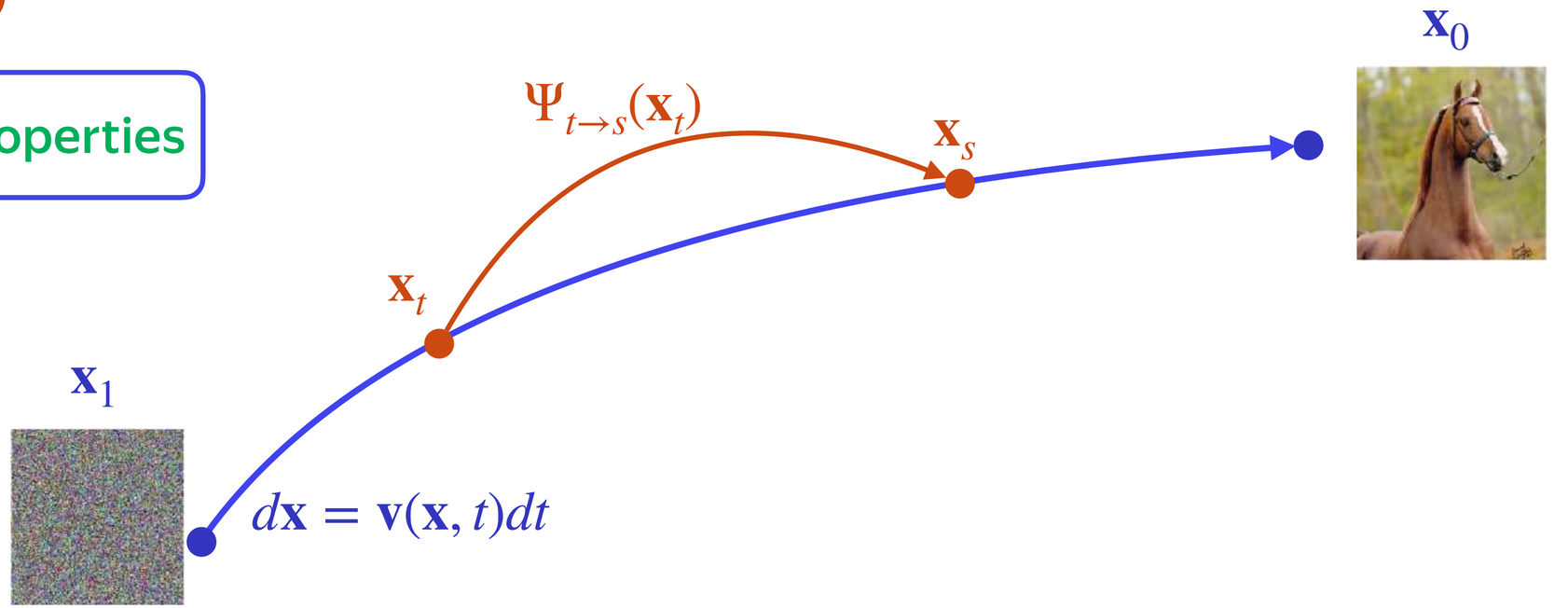
Goal: train $F_\theta(\mathbf{x}_t, t, s) \approx \Psi_{t\to s}(\mathbf{x}_t)$

How? Design loss via integral properties



Knowledge Distillation

Solve entire ODE: $\Psi_{1\to 0}(\mathbf{x}_t) = \mathbf{x}_1 + \int_1^0 \mathbf{v}(\mathbf{x}, u)\, du$

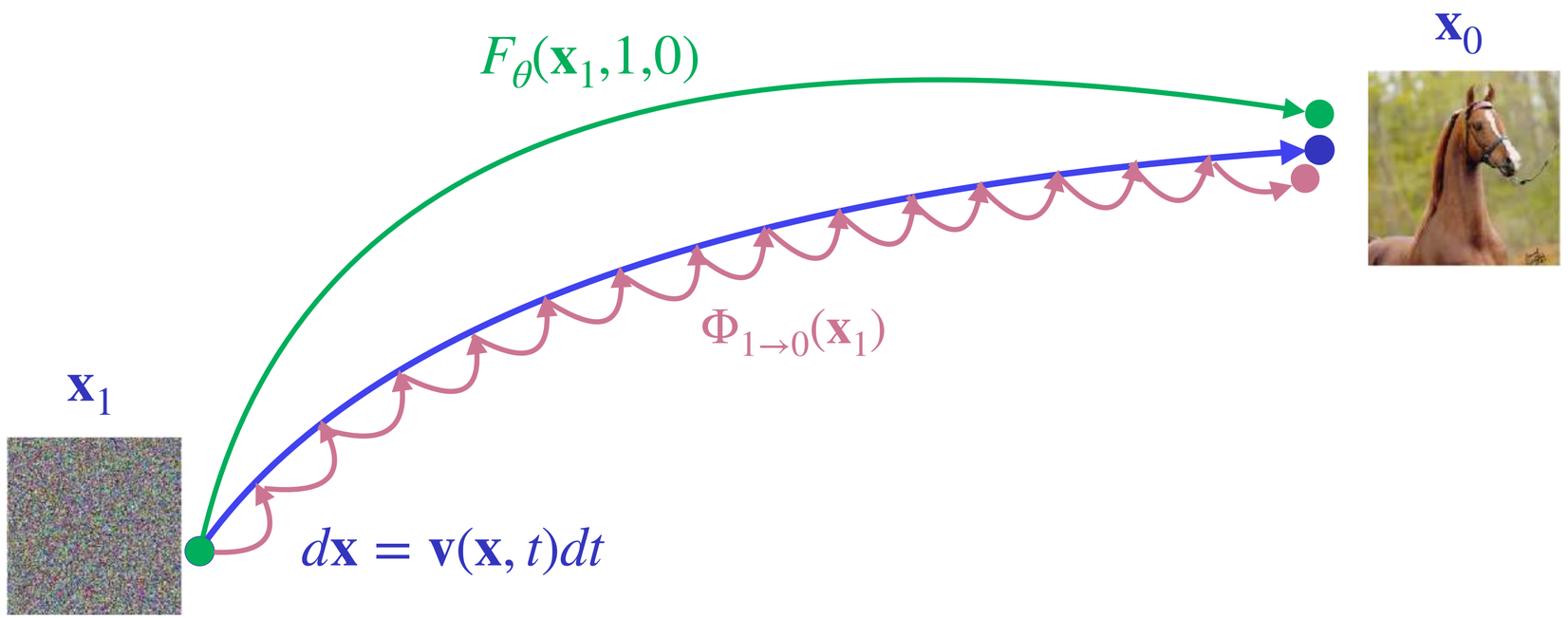
Goal: train $F_\theta(\mathbf{x}_1, 1, 0) \approx \Psi_{t\to 0}(\mathbf{x}_1)$

How? $\|F_\theta(\mathbf{x}_1,1,0) - \Phi_{1\to 0}(\mathbf{x}_1)\|_2^2$

Approximate teacher solutions

$\Phi_{1\to 0}(\mathbf{x}_1)$ — “oracle” DM

Sample with the pretrained DM



Consistency Models

Fix a zero boundary: $\Psi_{t\to 0}(\mathbf{x}_t) = \mathbf{x}_t + \int_t^0 \mathbf{v}(\mathbf{x}, u)\, du$

Goal: train $F_\theta(\mathbf{x}_t, t, 0) \approx \Psi_{t\to 0}(\mathbf{x}_t)$

How? $\| F_\theta(\mathbf{x}_t, t, 0) - sg\left(F_\theta(\mathbf{x}_r, r, 0)\right) \|_2^2$

*$sg(\cdot)$ — stopgrad

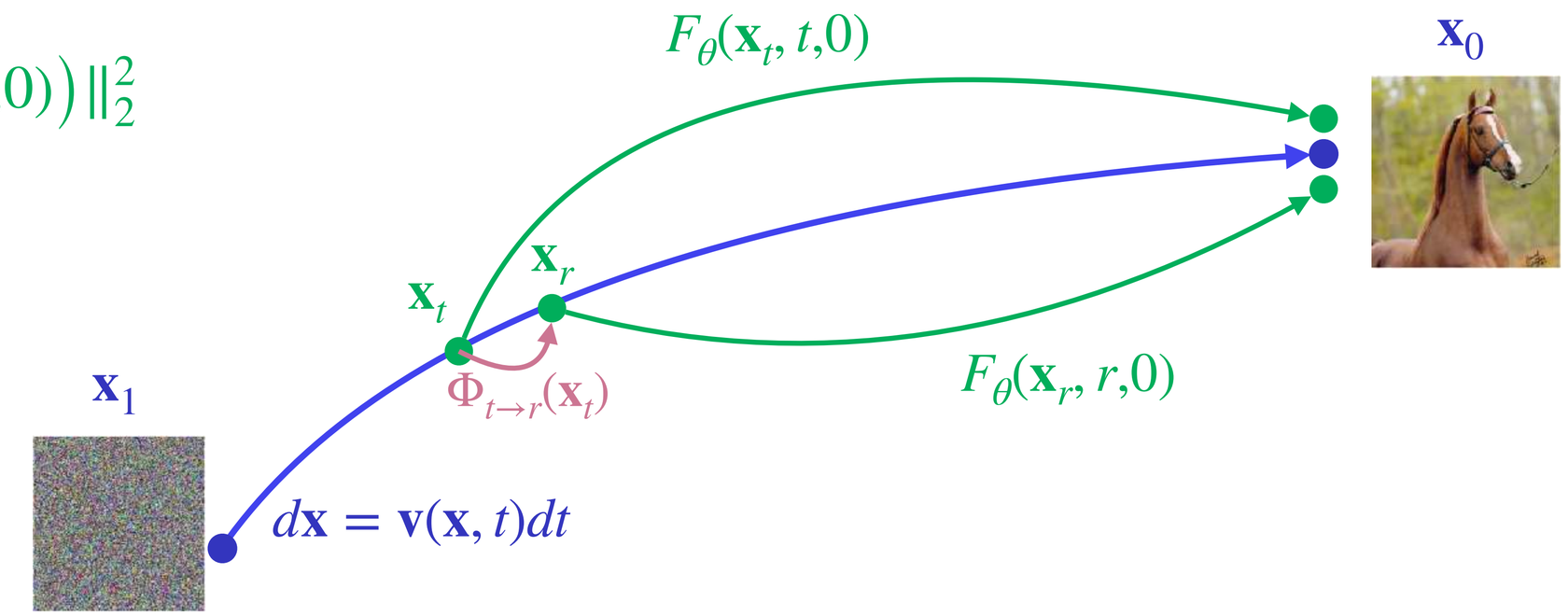
Self-consistency

$\Phi_{t\to r}(\mathbf{x}_t)$ — "oracle" DM

$[t, r]$ is small

⬇

one step using teacher or analytical score



Multi-boundary Consistency Models

K **boundaries** $s = \{s_0, \ldots, s_K\}$: $\Psi_{t \to s_k}(\mathbf{x}_t) = \mathbf{x}_t + \int_t^{s_k} \mathbf{v}(\mathbf{x}, u)\, du$

Goal: train $F_\theta(\mathbf{x}_t, t, s_k) \approx \Psi_{t \to s_k}(\mathbf{x}_t)$

How? $\| F_\theta(\mathbf{x}_t, t, s_k) - sg\left(F_\theta(\mathbf{x}_r, r, s_k)\right) \|_2^2$

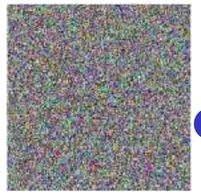
Self-consistency for K boundaries

$\Phi_{t \to r}(\mathbf{x}_t)$ — "oracle" DM

$[t, r]$ is small

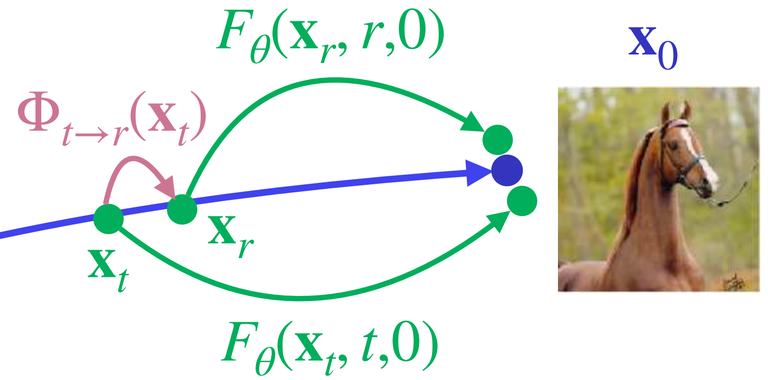
⬇

one step using teacher or analytical score



$\mathbf{x}_1$

$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$



Example: $s = \{0\}$

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Goal: train $F_\theta(\mathbf{x}_t, t, s_k) \approx \Psi_{t \to s_k}(\mathbf{x}_t)$

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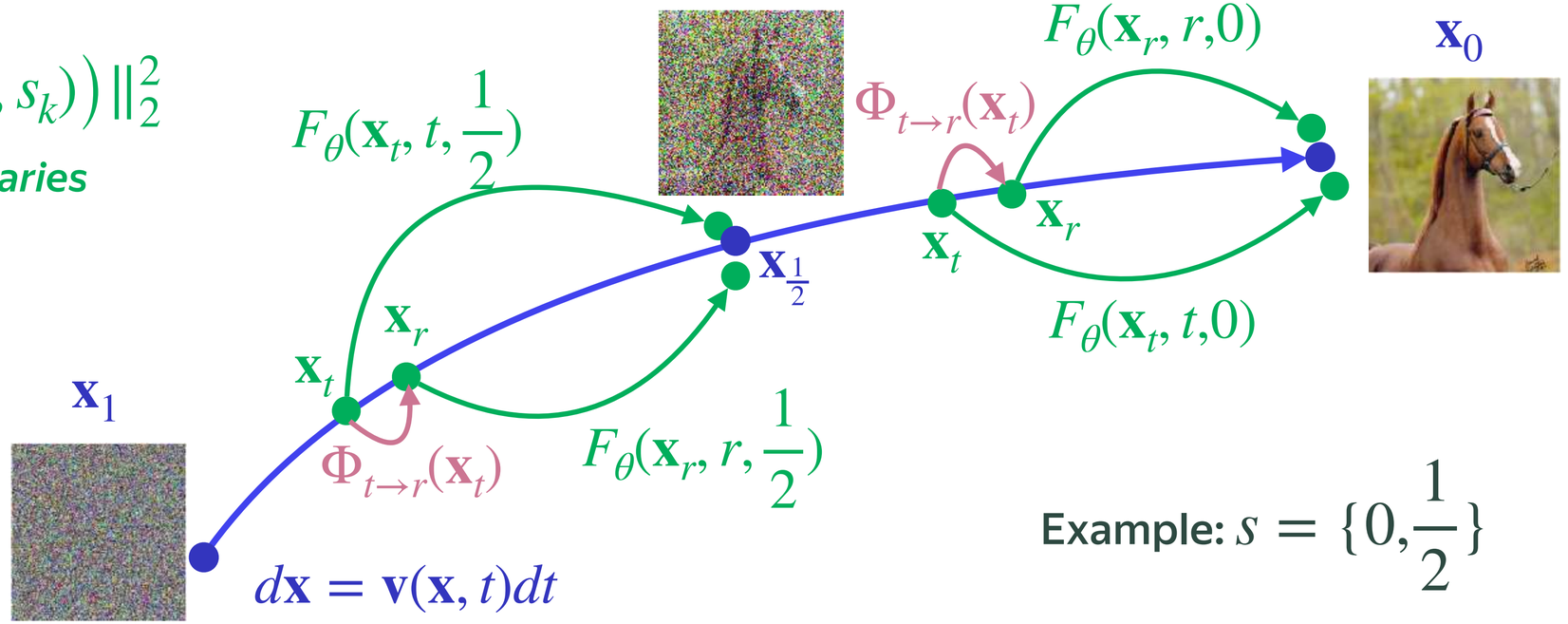
Self-consistency for K boundaries

$\Phi_{t \to r}(\mathbf{x}_t)$ — "oracle" DM

$[t, r]$ is small

⬇

one step using teacher or analytical score



Example: $s = \{0, \frac{1}{2}\}$

Consistency Trajectory Models

Continuous $s \in (1,0]$: $\Psi_{t\to s}(\mathbf{x}_t) = \mathbf{x}_t + \int_t^s \mathbf{v}(\mathbf{x}, u)\, du$

Goal: train $F_\theta(\mathbf{x}_t, t, s) \approx \Psi_{t\to s}(\mathbf{x}_t)$

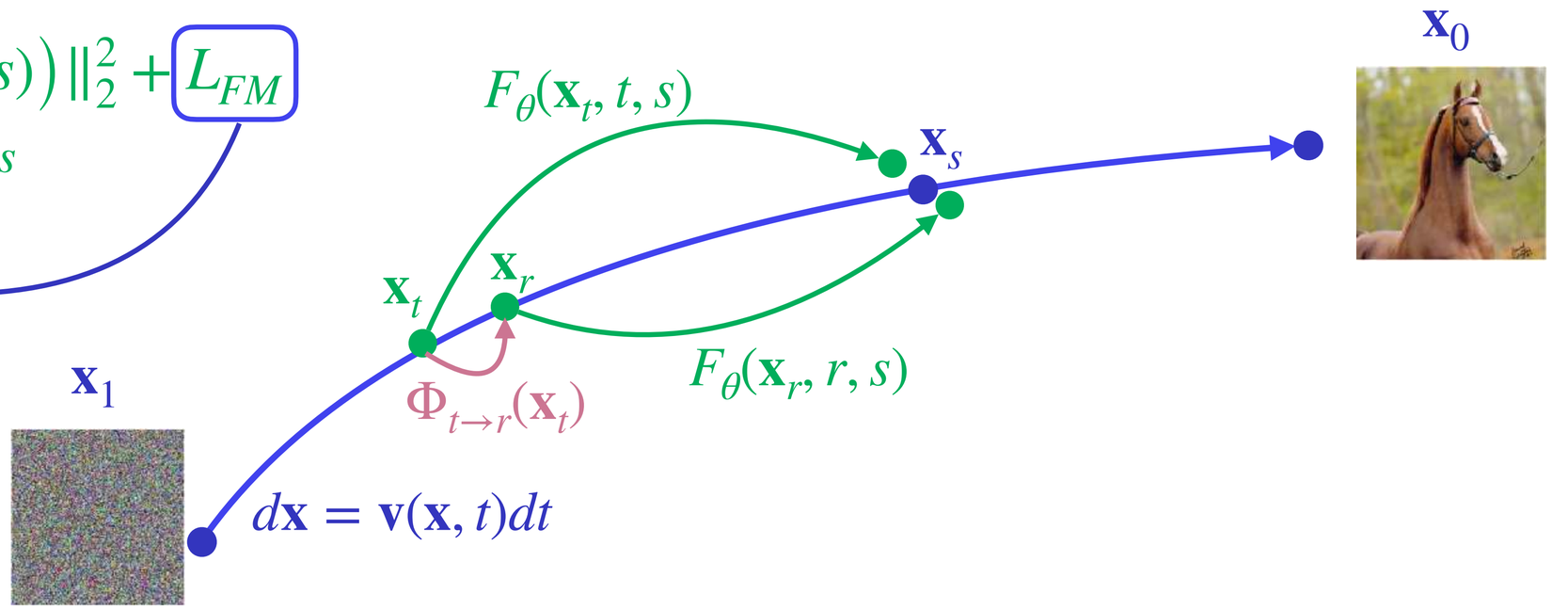
How? $\| F_\theta(\mathbf{x}_t, t, s) - sg\big(F_\theta(\mathbf{x}_r, r, s)\big) \|_2^2 + L_{FM}$

Self-consistency for arbitrary s

$F_\theta(\mathbf{x}_t, t, s) = \mathbf{x}_t + (s-t)\mathbf{v}_\theta(\mathbf{x}_t, t)$

$\Phi_{t\to r}(\mathbf{x}_t)$ — "oracle" DM

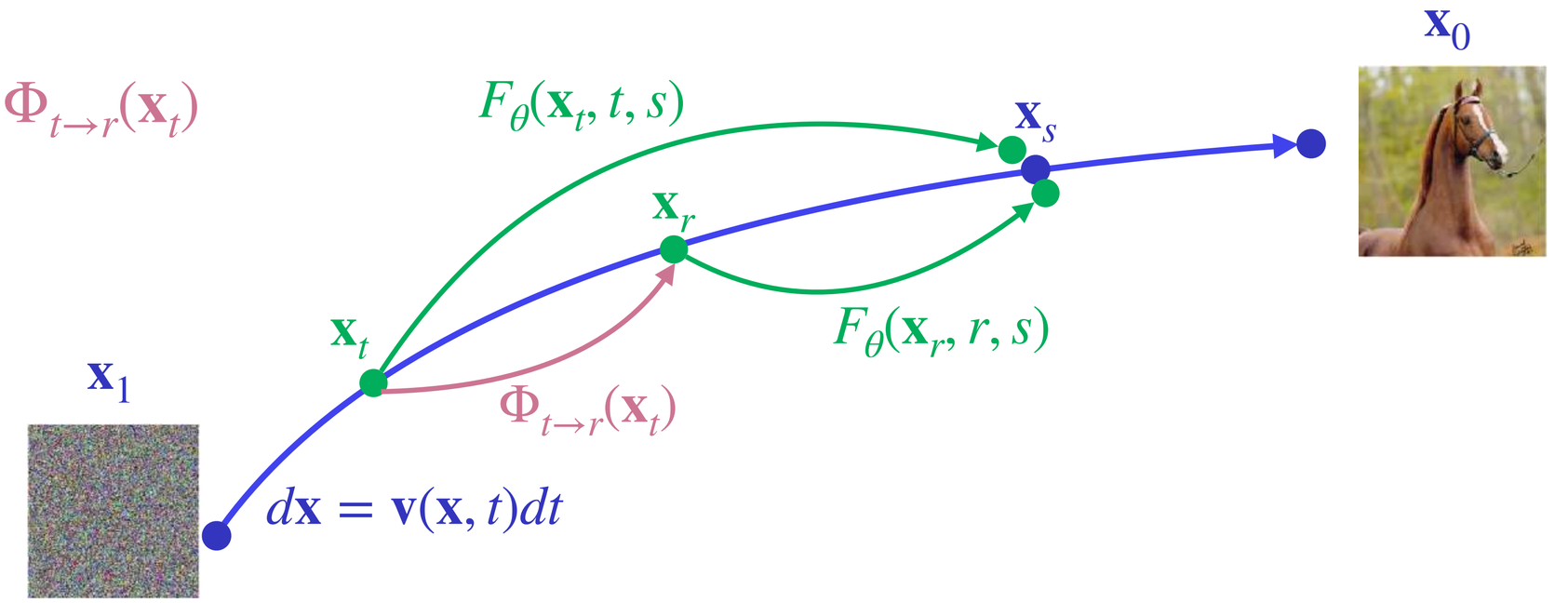
one step using teacher or $\mathbf{v}_\theta(\mathbf{x}_t, t)$



General Consistency Trajectory Models

$[t, r]$ can be arbitrary: small or large

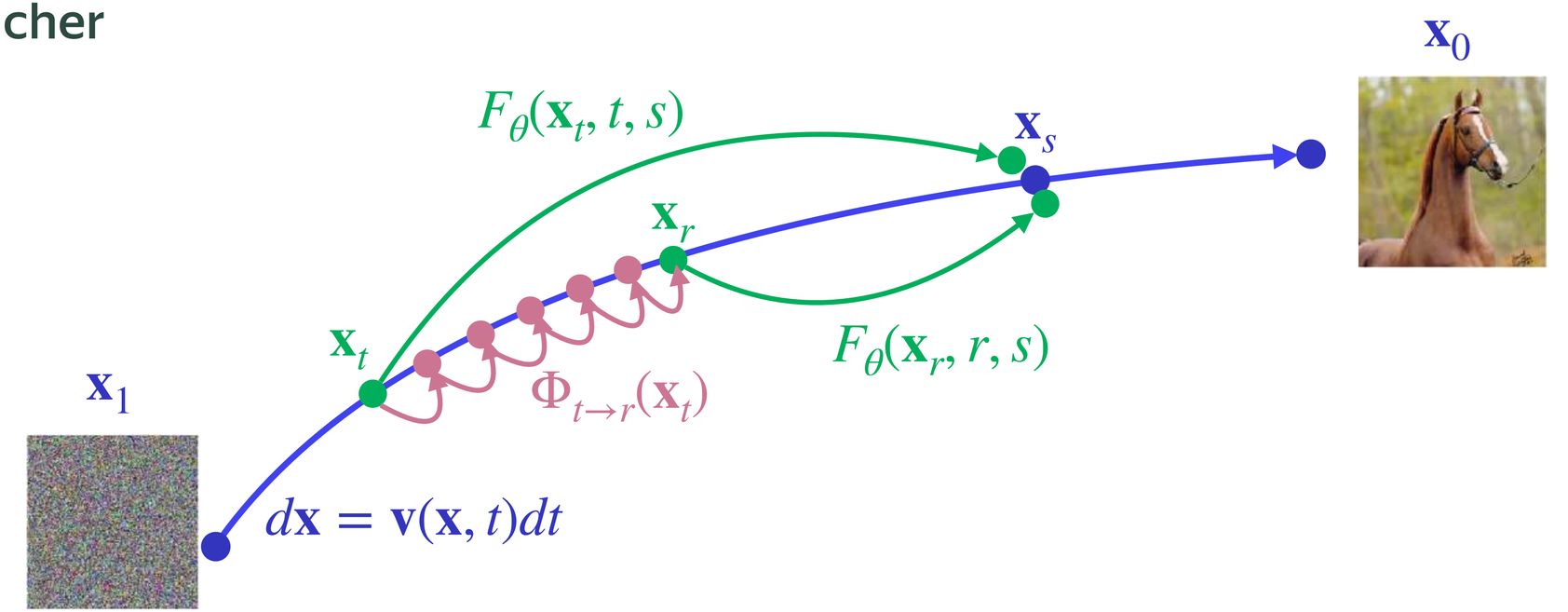
There are general choices for $\Phi_{t\to r}(\mathbf{x}_t)$



General Consistency Trajectory Models

“Oracle” diffusion choices

$\Phi_{t\to r}(\mathbf{x}_t)$ — multi-step teacher

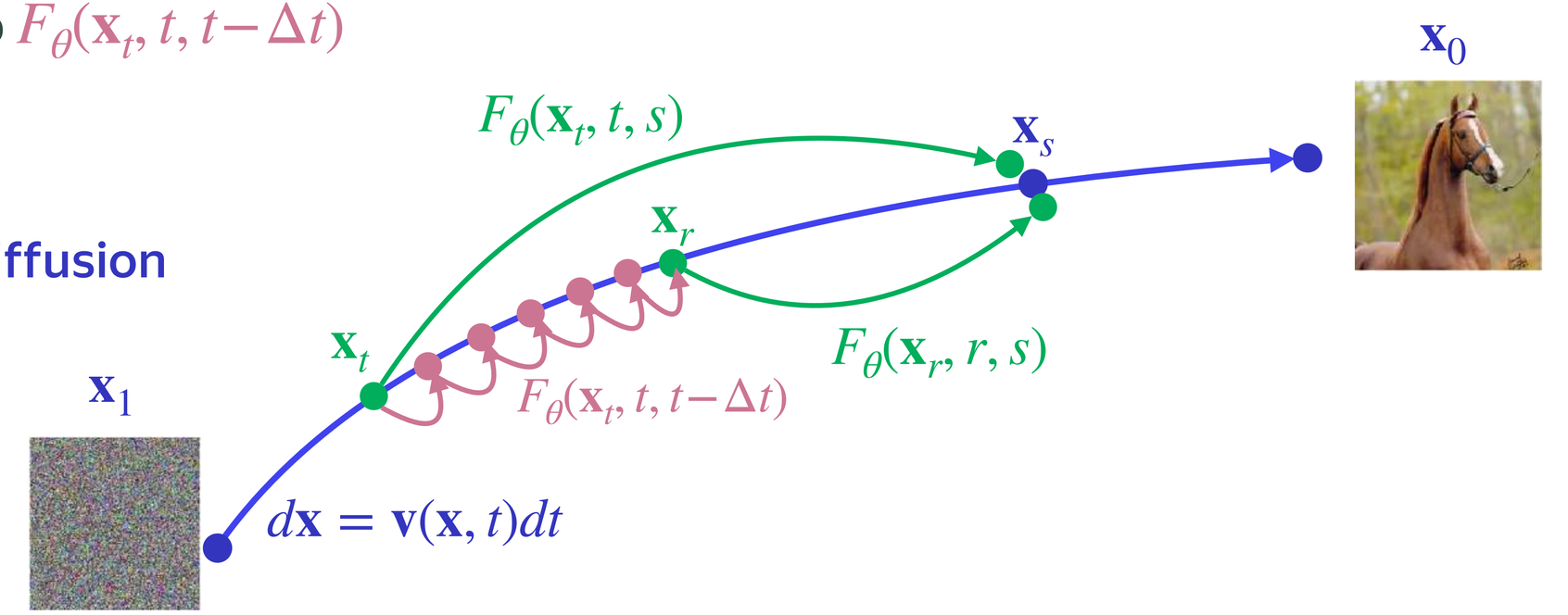


General Consistency Trajectory Models

“Oracle” diffusion choices

$\Phi_{t\to r}(\mathbf{x}_t)$ — multi-step flow-map $F_\theta(\mathbf{x}_t, t, t-\Delta t)$

No need in pretrained diffusion

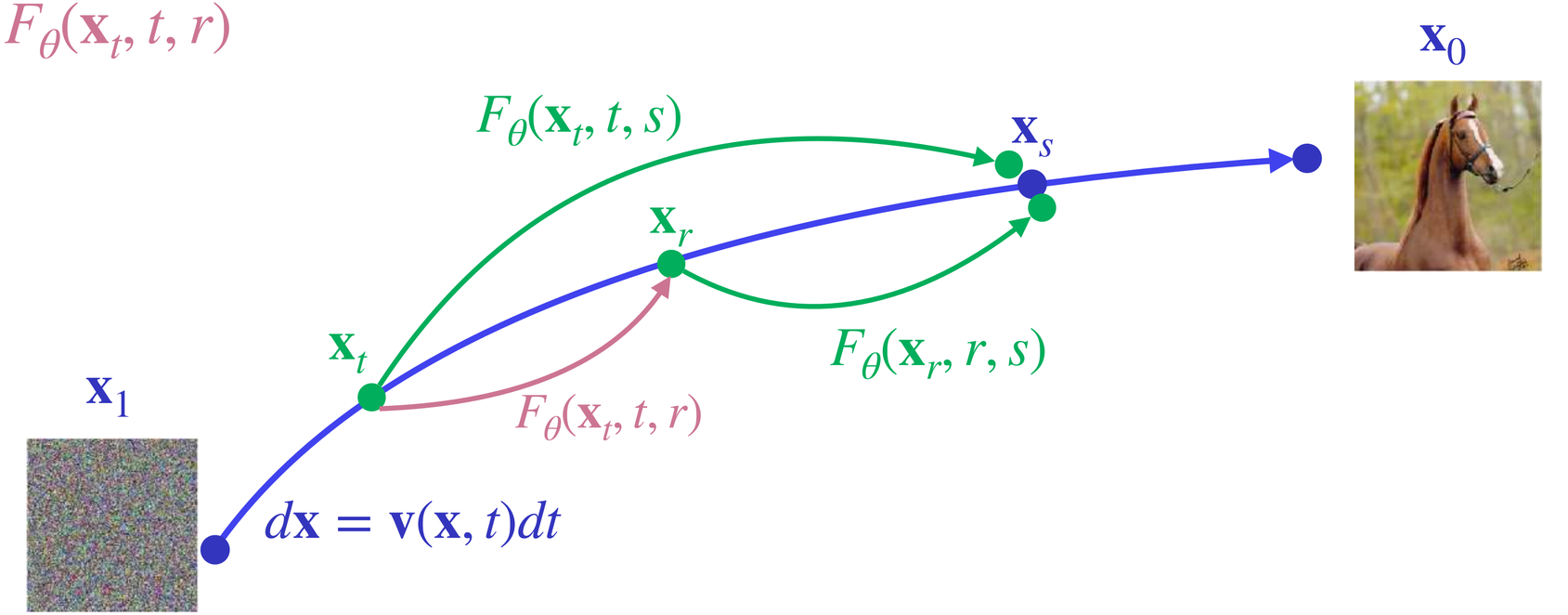


General Consistency Trajectory Models

“Oracle” diffusion choices

$\Phi_{t\to r}(\mathbf{x}_t)$ — one-step flow-map $F_\theta(\mathbf{x}_t, t, r)$

Shortcut Models



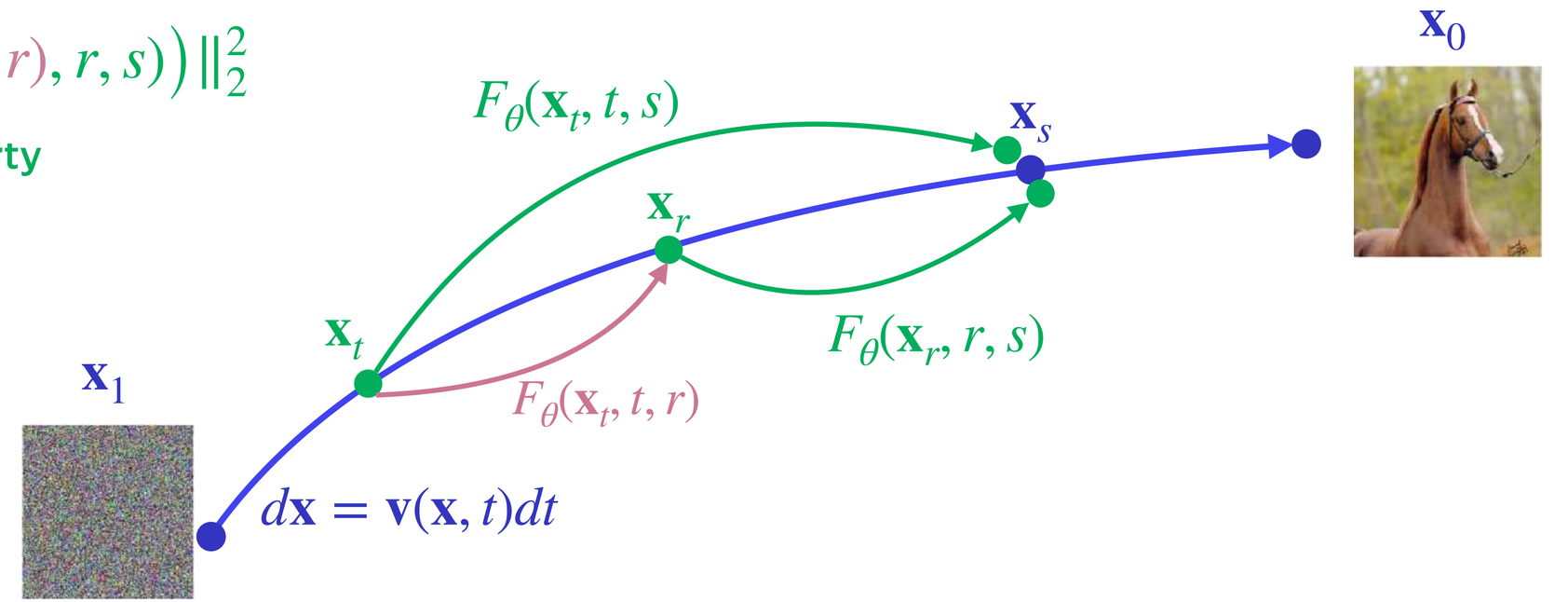
Shortcut Models

Interval additivity property

$$\int_t^s \mathbf{v}(\mathbf{x}, u)\, du = \int_t^r \mathbf{v}(\mathbf{x}, u)\, du + \int_r^s \mathbf{v}(\mathbf{x}, u)\, du$$

$$\| F_\theta(\mathbf{x}_t, t, s) - sg\left(F_\theta(F_\theta(\mathbf{x}_t, t, r), r, s)\right) \|_2^2$$

Interval additivity property



Mean Flow

$$F_\theta(\mathbf{x}_t, t, s) \approx \frac{1}{s-t}\int_t^s \mathbf{v}(\mathbf{x}, u)\, du$$

Mean Flow

$$F_\theta(\mathbf{x}_t, t, s) \approx \frac{1}{s-t}\int_t^s \mathbf{v}(\mathbf{x}, u)\, du \Rightarrow \Psi_{t\to s}(\mathbf{x}_t) \approx \mathbf{x}_t + (s-t)F_\theta(\mathbf{x}_t, t, s)$$

Mean Flow

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$$(s-t)F_\theta(\mathbf{x}_t, t, s) = \int_t^s \mathbf{v}(\mathbf{x}, u)\, du$$

$$\downarrow \frac{d}{dt}$$

Mean Flow Identity

$$F_\theta(\mathbf{x}_t, t, s) = \mathbf{v}(\mathbf{x}_t, t) - (s-t)\frac{d}{dt}F_\theta(\mathbf{x}_t, t, s)$$

Avg. velocity

Inst. velocity

Time derivative

Mean Flow

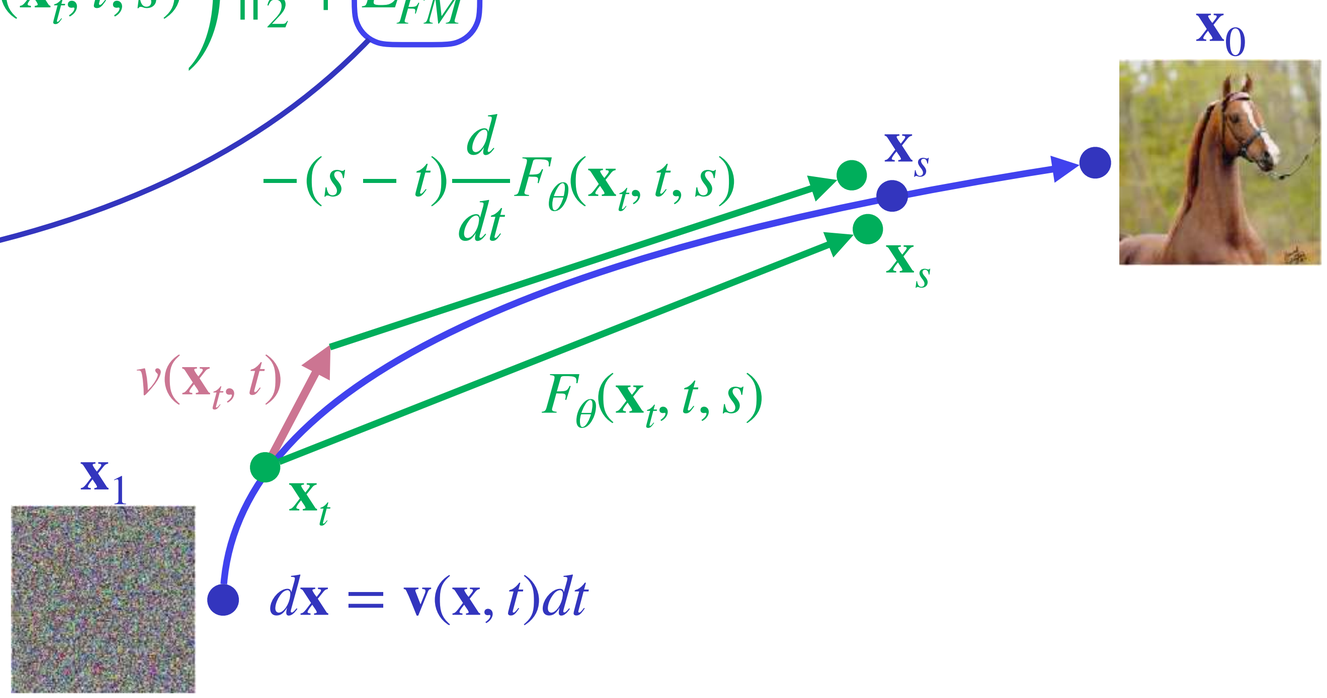
Goal: train $F_\theta(\mathbf{x}_t, t, s) \approx \frac{1}{s-t}\int_t^s \mathbf{v}(\mathbf{x}, u)\, du$

How? $\|F_\theta(\mathbf{x}_t, t, s) - sg\left(\mathbf{v}(\mathbf{x}_t, t) - (s-t)\frac{d}{dt}F_\theta(\mathbf{x}_t, t, s)\right)\|_2^2 + L_{FM}$

Mean Flow Identity

$F_\theta(\mathbf{x}_t, t, t) = \mathbf{v}_\theta(\mathbf{x}_t, t)$

$\mathbf{v}(\mathbf{x}_t, t)$ — $\dot{\mathbf{x}}_t$ or DM prediction $\mathbf{v}_\phi(\mathbf{x}_t, t)$



Takeaways

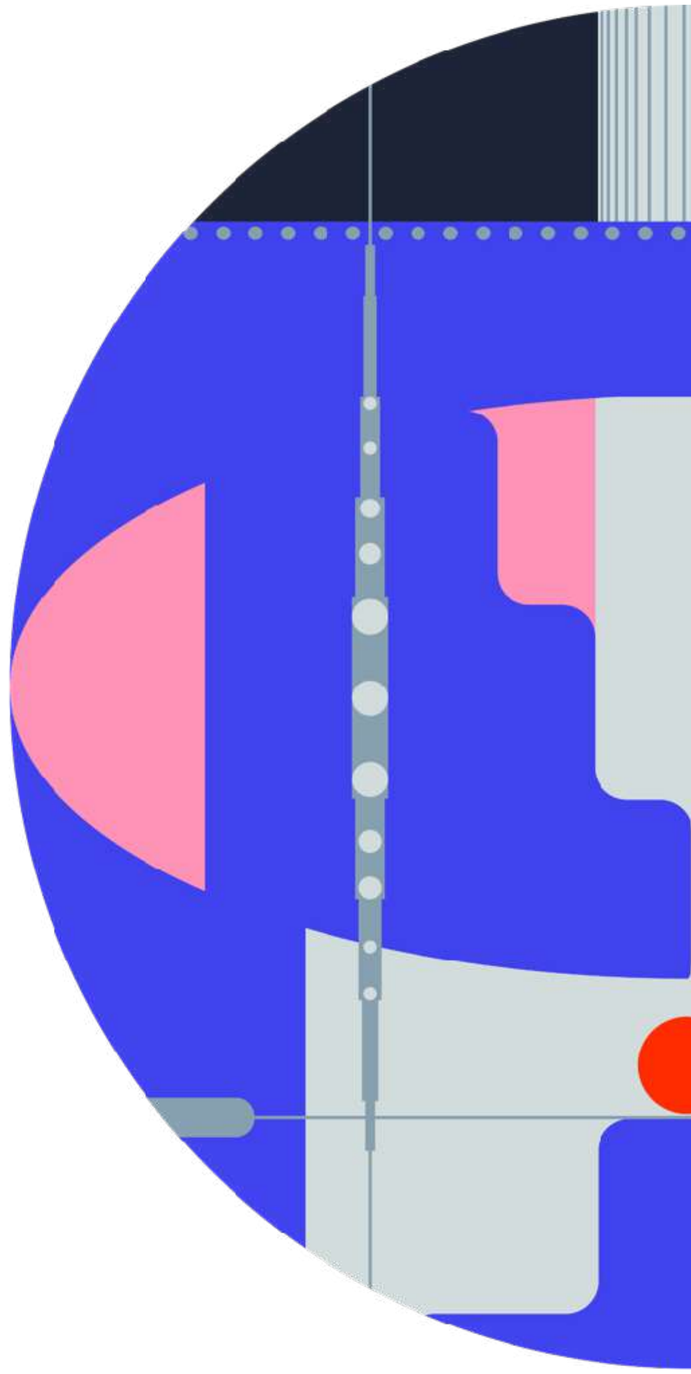
KD — the most naïve approach. Still useful for regularization, or for warming up few-step models

Multi-boundary (-step) CM — a strong and practical method, particularly effective for 6–8 steps.

CTM / Shortcut Model — elegant and productive ideas. I did not see large-scale applications

Continuous CM / MeanFlow — elegant and promising formulations, but too complex in practice

Preliminaries



Notation

Generator $G_\theta(\mathbf{z})$ produces $p_{fake}(\mathbf{x})$, where $\mathbf{z} \sim \mathcal{N}(0, I)$

Data distribution $p_{real}(\mathbf{x})$

Forward process: $q(\mathbf{x}_t \mid \mathbf{x}) = \mathcal{N}(\alpha_t \mathbf{x}, \sigma_t^2 I)$

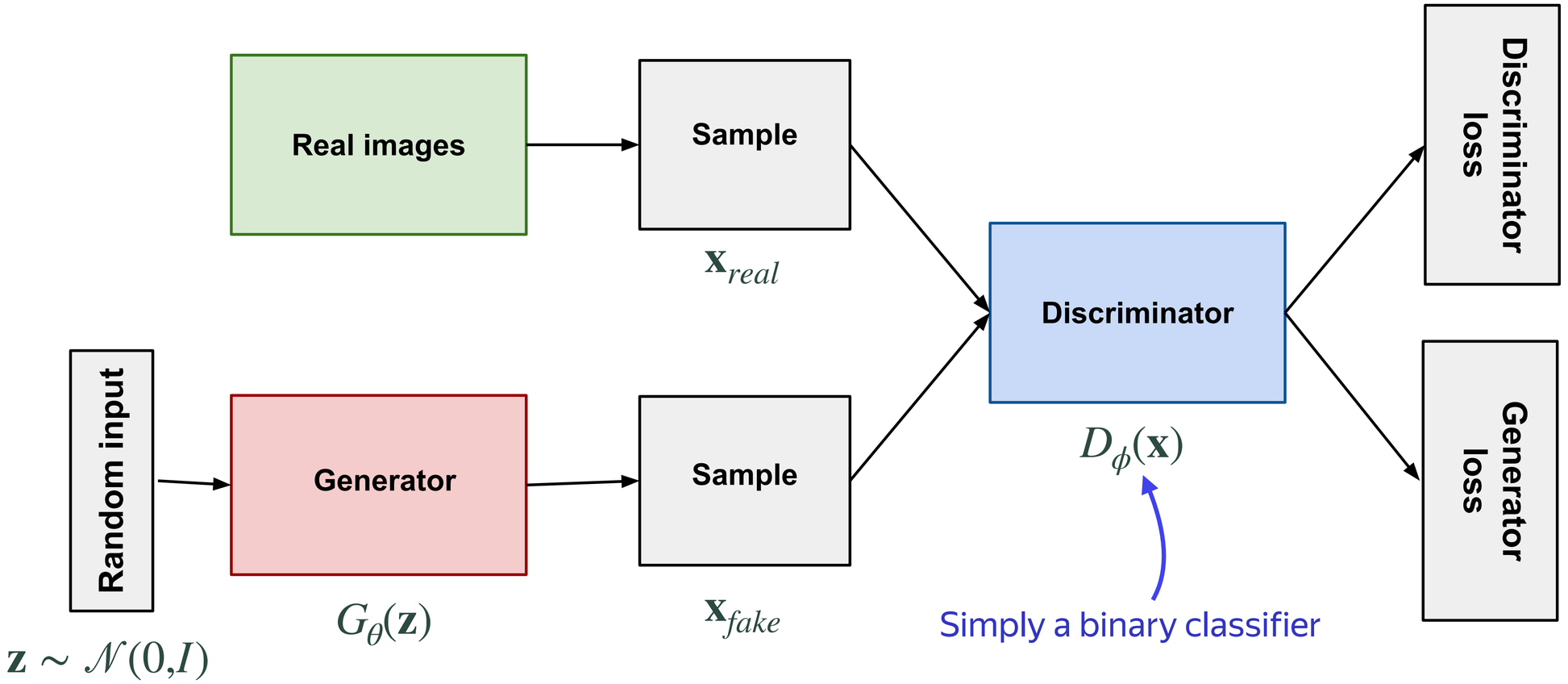
Pretrained diffusion model (DM) $\epsilon_\psi(\mathbf{x}_t, t)$

Tweedie's formula

$$\nabla_{\mathbf{x}_t} \log p_{real}(\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) = \mathbb{E}\left[\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t \mid \mathbf{x}) \mid \mathbf{x}_t\right] \approx \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t \mid \mathbf{x}) = -\frac{\mathbf{x}_t - \alpha_t \mathbf{x}}{\sigma_t^2} = -\frac{\epsilon}{\sigma_t}$$

Recap of GANs

Generator $G_\theta(\mathbf{z}) \rightarrow \mathbf{x}$ tries to fool the discriminator $D_\phi(\mathbf{x})$



Recap of GANs

Standard Min-Max loss

Discriminator loss: $$\max_{\phi} \left[\mathop{\mathbb{E}}_{\mathbf{x} \sim p_{real}} \log D_{\phi}(\mathbf{x}) + \mathop{\mathbb{E}}_{\mathbf{z} \sim \mathcal{N}(0,I)} \log(1 - D_{\phi}(G_{\theta}(\mathbf{z}))) \right]$$

Generator loss: $$\min_{\theta} \mathop{\mathbb{E}}_{\mathbf{z} \sim \mathcal{N}(0,I)} \log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))$$

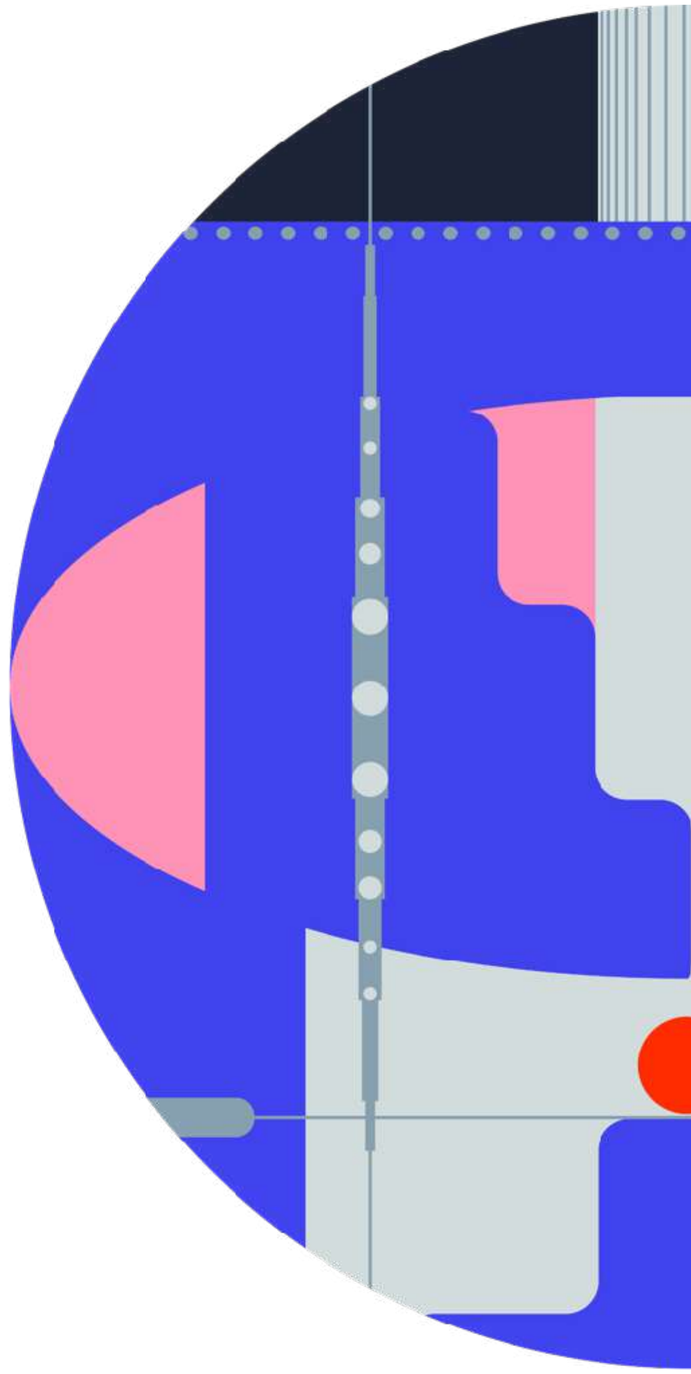
Non-saturating generator loss: $$\min_{\theta} \mathop{\mathbb{E}}_{\mathbf{z} \sim \mathcal{N}(0,I)} - \log D_{\phi}(G_{\theta}(\mathbf{z}))$$

Recap of GANs

GANs are powerful **single step** generative models

but struggle to achieve s.o.t.a. performance w/o some help, e.g., **diffusion**

Distribution Matching Distillation



Training objective

Let’s consider reverse KL objective

$$D_{KL}(p_{fake} \| p_{real}) = \mathop{\mathbb{E}}_{p_{fake}(\mathbf{x})} \left[\log p_{fake}(\mathbf{x}) - \log p_{real}(\mathbf{x}) \right]$$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Training objective

Let's consider reverse KL objective

Reparameterization trick

$$D_{KL}(p_{fake} \| p_{real}) = \mathop{\mathbb{E}}_{p_{fake}(\mathbf{x})} \left[\log p_{fake}(\mathbf{x}) - \log p_{real}(\mathbf{x}) \right] = \mathop{\mathbb{E}}_{\mathbf{z} \sim \mathcal{N}(0,I)} \left[\log p_{fake}(G_\theta(\mathbf{z})) - \log p_{real}(G_\theta(\mathbf{z})) \right]$$

$\log p_{fake}(\mathbf{x})$ and $\log p_{real}(\mathbf{x})$ are unavailable

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Training objective

$$D_{KL}(p_{fake} \| p_{real}) = \mathop{\mathbb{E}}_{z \sim \mathcal{N}(0,I)} \left[\log p_{fake}(G_\theta(\mathbf{z})) - \log p_{real}(G_\theta(\mathbf{z})) \right]$$

$$\nabla_\theta D_{KL}(p_{fake} \| p_{real}) = \mathop{\mathbb{E}}_{z \sim \mathcal{N}(0,I)} \left[\nabla_\theta \log p_{fake}(G_\theta(\mathbf{z})) - \nabla_\theta \log p_{real}(G_\theta(\mathbf{z})) \right] =$$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Training objective

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$$\nabla_\theta D_{KL}(p_{fake} \| p_{real}) = \mathop{\mathbb{E}}_{z \sim \mathcal{N}(0,I)} \left[\nabla_\theta \log p_{fake}(G_\theta(\mathbf{z})) - \nabla_\theta \log p_{real}(G_\theta(\mathbf{z})) \right] = \text{Chain rule}$$

$$= \mathop{\mathbb{E}}_{z \sim \mathcal{N}(0,I)} \left[(\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})) \nabla_\theta G_\theta(\mathbf{z}) \right], \quad \text{where } \mathbf{x} = G_\theta(\mathbf{z})$$

Can we compute $\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x})$ and $\nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$?

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Training objective

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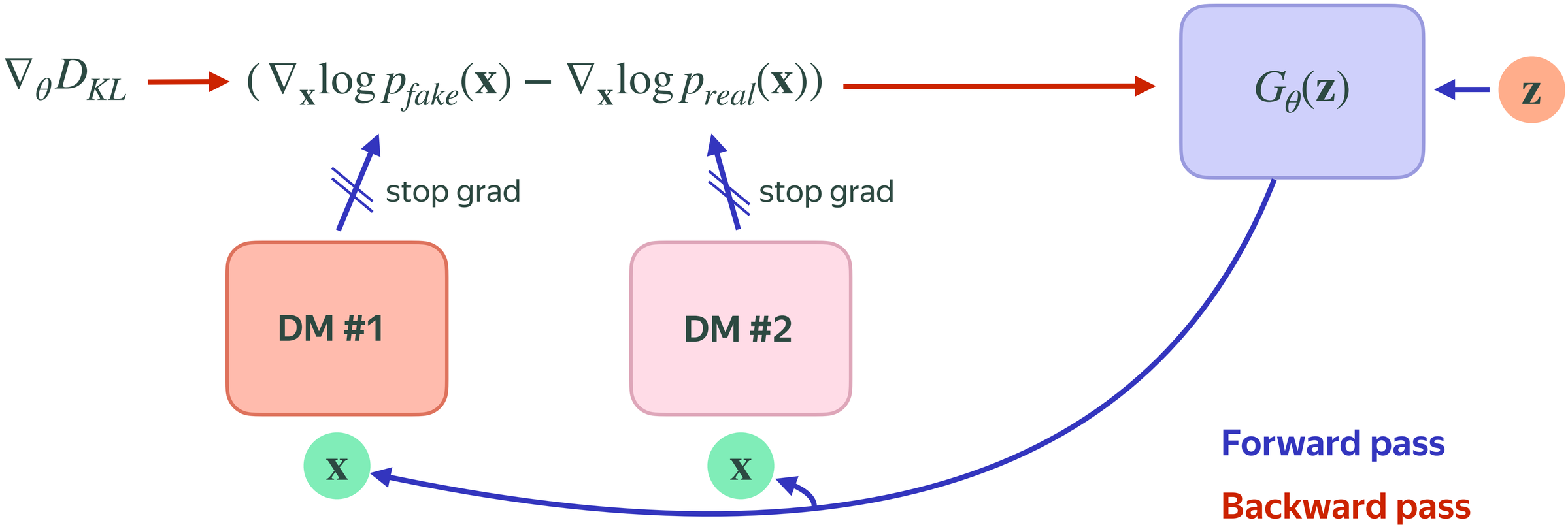
$$= \mathop{\mathbb{E}}_{z \sim \mathcal{N}(0,I)} \left[(\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})) \nabla_\theta G_\theta(\mathbf{z}) \right], \quad \text{where } \mathbf{x} = G_\theta(\mathbf{z})$$

Use DM forward passes to estimate $\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x})$ and $\nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Generator update



Key idea

Pretrained DM estimates $\nabla_{\mathbf{x}_t} \log p_{real}(\mathbf{x}_t) \approx s^{\psi}_{real}(\mathbf{x}_t, t) = -\frac{\epsilon_{\psi}(\mathbf{x}_t, t)}{\sigma_t}$

For $t = 0$, we have $s^{\psi}_{real}(\mathbf{x}_0, 0) \approx \nabla_{\mathbf{x}_0} \log p_{real}(\mathbf{x}_0) \approx \nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

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For $t = 0$, we have $s^{\psi}_{real}(\mathbf{x}_0, 0) \approx \nabla_{\mathbf{x}_0} \log p_{real}(\mathbf{x}_0) \approx \nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$

Important: DM is pretrained on $p_{real}(\mathbf{x})$ → it can estimate only $\nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$

How to estimate $\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x})$?

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Key idea

Let's train a **new** DM $s_\phi(\mathbf{x}_t, t)$ on $p_{fake}(\mathbf{x})$, produced by **current** G_θ

For $t = 0$, we have $s^{\phi}_{fake}(\mathbf{x}_0, 0) \approx \nabla_{\mathbf{x}_0} \log p_{fake}(\mathbf{x}_0) \approx \nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x})$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

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Problem: $p_{fake}(\mathbf{x})$ changes after each G_θ update

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Problem: $p_{fake}(\mathbf{x})$ changes after each G_θ update

Solution: re-train $s^\phi_{fake}(\mathbf{x}_t, t)$ after each G_θ update

For efficiency, a single update per training iteration is okay

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Algorithm

1. Update step for G_θ

$$\nabla_\theta D_{KL} \simeq \mathop{\mathbb{E}}_{\mathbf{x}\sim G_\theta(\mathbf{z})} \left[\left(s^{\phi}_{fake}(\mathbf{x},0) - s^{\psi}_{real}(\mathbf{x},0) \right) \nabla_\theta G_\theta(\mathbf{z}) \right]$$

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2. Update step for fake DM ϵ_ϕ

Typical DM training step

$$\nabla_\phi \mathop{\mathbb{E}}_{\mathbf{z},t,\mathbf{x},\mathbf{x}_t} \| s^{\phi}_{fake}(q(\mathbf{x}_t \mid \mathbf{x}), t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t \mid \mathbf{x}) \|_2^2 = \nabla_\phi \mathop{\mathbb{E}}_{\mathbf{z},t,\mathbf{x},\mathbf{x}_t} \| \epsilon_\phi(q(\mathbf{x}_t \mid \mathbf{x}), t) - \epsilon \|_2^2$$

What is a problem here?

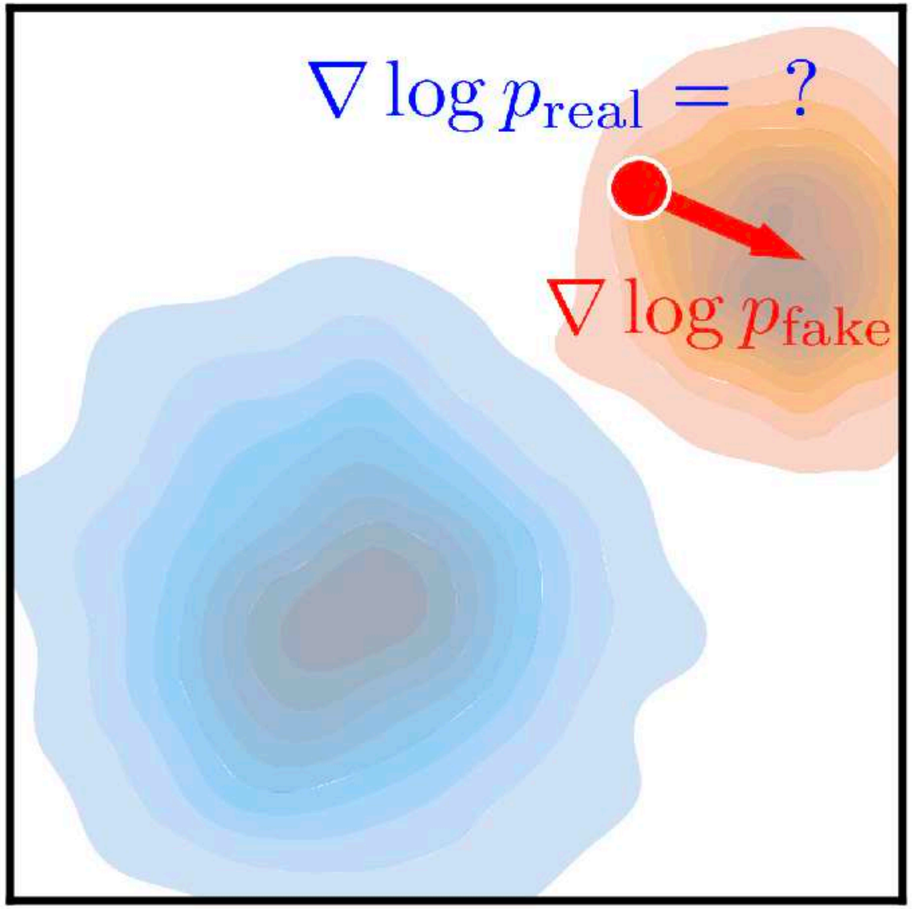
T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Undefined scores

G_θ produces low quality samples at the beginning

Pretrained DM observed nearly real samples for $t = 0$

$\rightarrow s^{\psi}_{real}(\mathbf{x},0)$ is inaccurate for $\mathbf{x} \sim p_{fake}(\mathbf{x})$

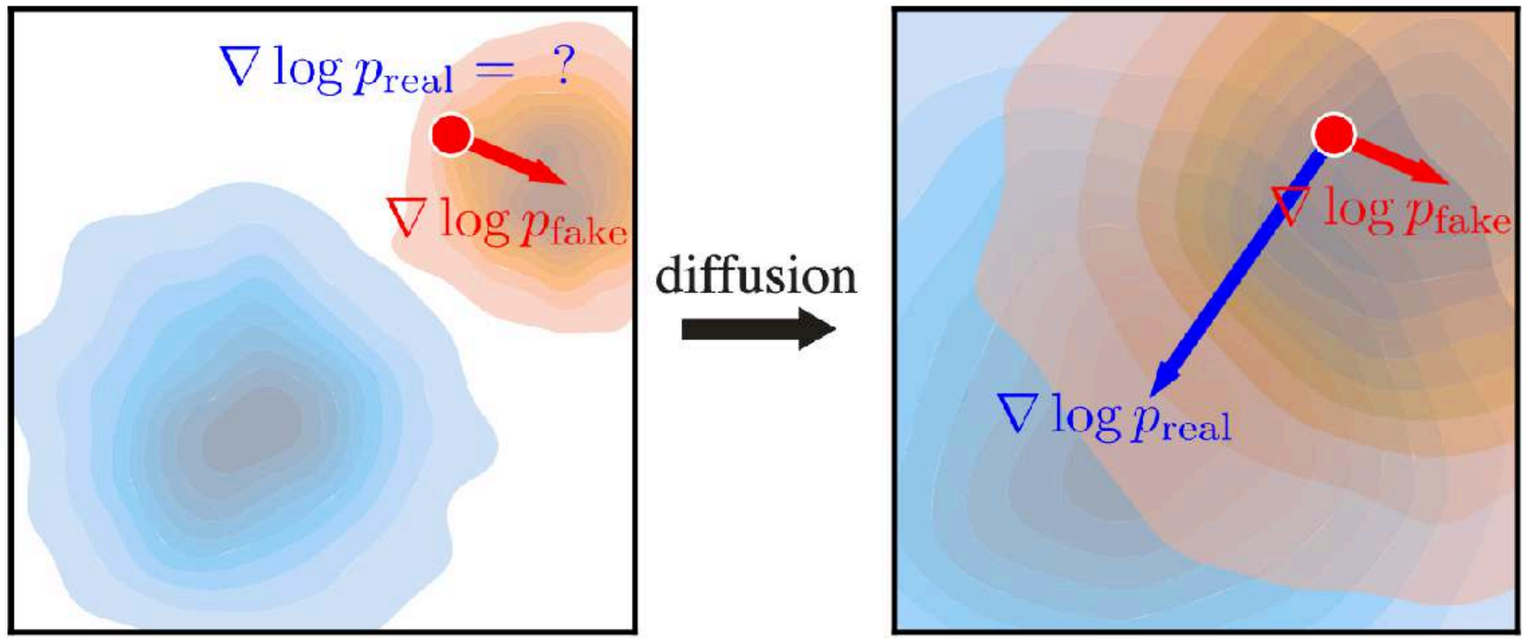


T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Undefined scores

Similar idea to the annealed sampling in denoising score matching

Let's use $t \gg 0$ to avoid undefined scores at the beginning, e.g., $t = T$



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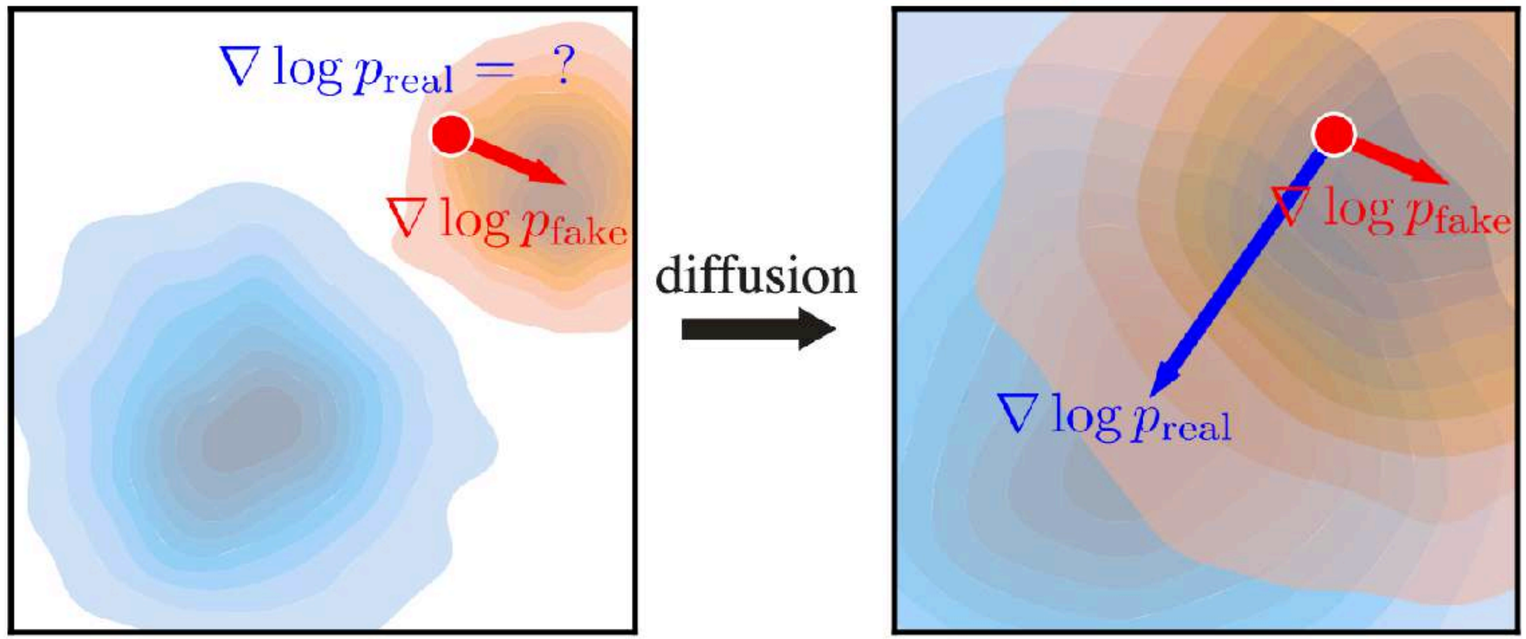
Undefined scores

Similar idea to the annealed sampling in denoising score matching

Let's use $t \gg 0$ to avoid undefined scores at the beginning, e.g., $t = T$

Gradually reduce t once G_θ gets better



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Overall algorithm

1. Update step for G_θ

$$\nabla_\theta D_{KL} \simeq \mathop{\mathbb{E}}_{\mathbf{z},t,\mathbf{x},\mathbf{x}_t} \left[\left(s^{\phi}_{fake}(\mathbf{x}_t, t) - s^{\psi}_{real}(\mathbf{x}_t, t) \right) \nabla_\theta G_\theta(\mathbf{z}) \right]$$

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

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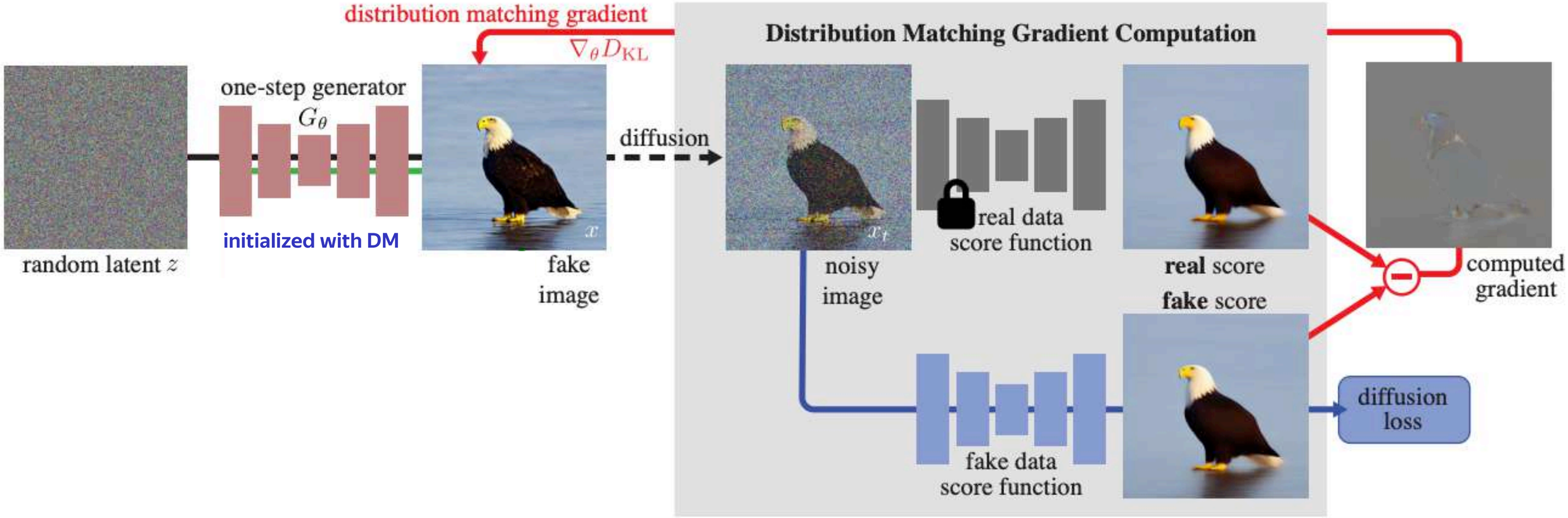
P.S.: pretty similar to GANs, where $s^{\phi}_{fake}(\mathbf{x}_t, t)$ and $s^{\psi}_{real}(\mathbf{x}_t, t))$ are “half-discriminators”

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Distribution Matching Distillation



T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

DMD loss

Practical detail: how to implement $\nabla_\theta D_{KL}$?

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

DMD loss

Practical detail: how to implement $\nabla_\theta D_{KL}$?

Surrogate loss: $$L = \frac{1}{2} \mathop{\mathbb{E}}_{\mathbf{z},t,\mathbf{x},\mathbf{x}_t} \left[\| G_\theta(\mathbf{z}) - sg\left(G_\theta(\mathbf{z}) - s^{\phi}_{fake}(\mathbf{x}_t, t) + s^{\psi}_{real}(\mathbf{x}_t, t) \right) \|_2^2 \right]$$

It can be shown that $\nabla_\theta L = \nabla_\theta D_{KL}$ via chain rule.

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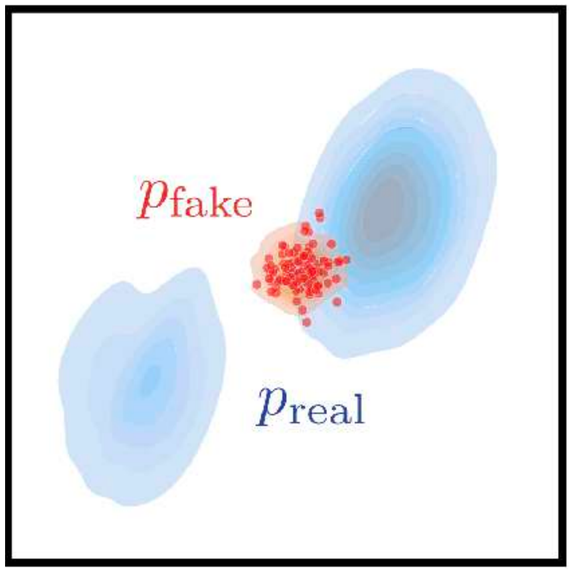
Reverse KL problem



Initial state

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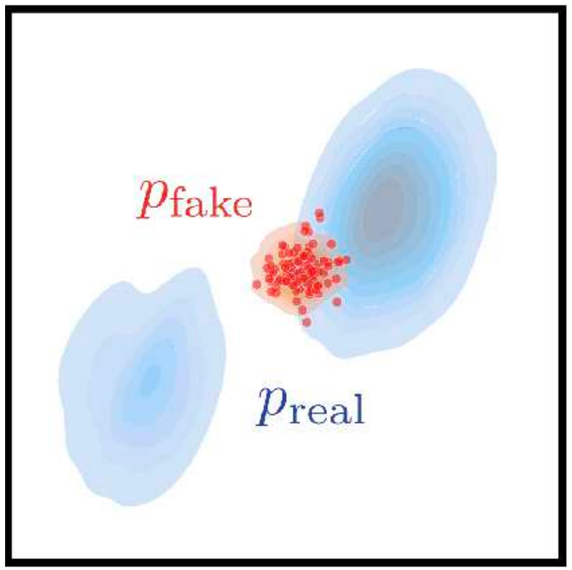
Reverse KL problem

$$\mathop{\mathbb{E}}_{\substack{z\sim\mathcal{N}(0;\mathbf{I})\\ x=G_\theta(z)}} -\big(\log\, p_{\text{real}}(x) - \log\, p_{\text{fake}}(x)\big)$$



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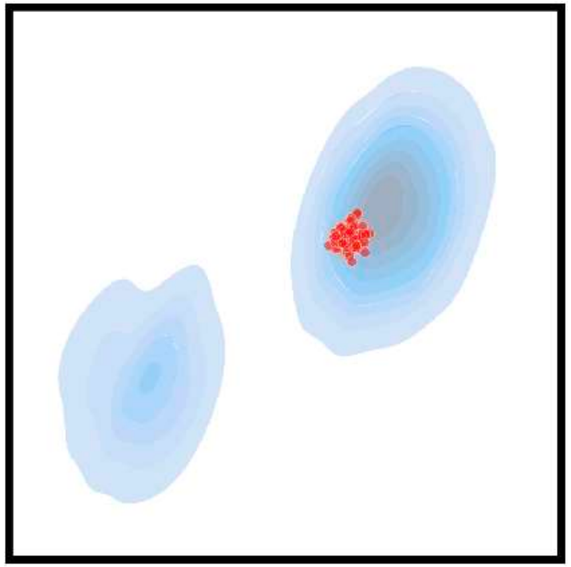
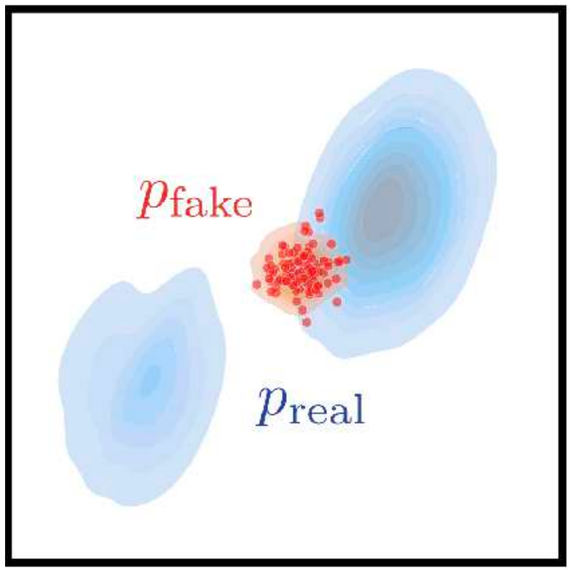


Real score only

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Reverse KL problem

$$\mathop{\mathbb{E}}_{\substack{z\sim\mathcal{N}(0;\mathbf{I})\\ x=G_\theta(z)}} -\big(\log\ p_{\text{real}}(x) - \log\ p_{\text{fake}}(x)\big)$$



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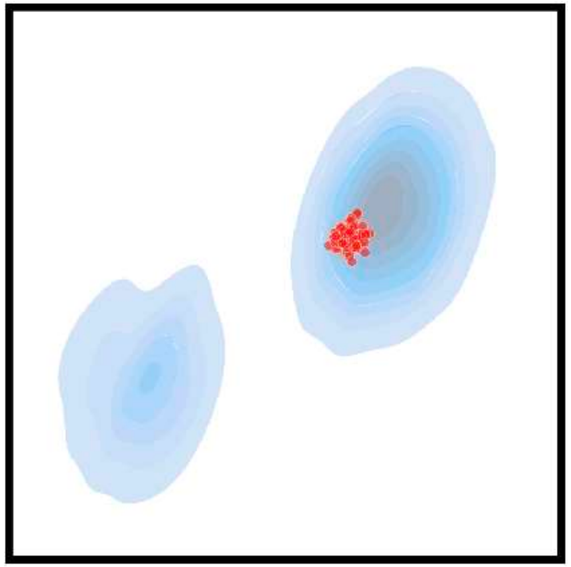

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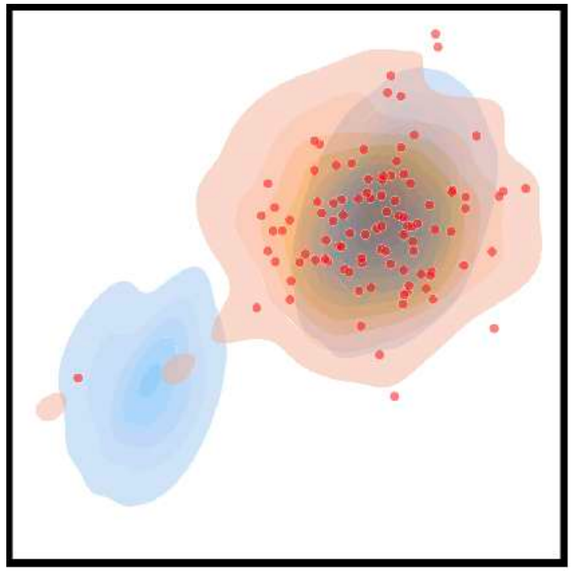

Real + fake scores

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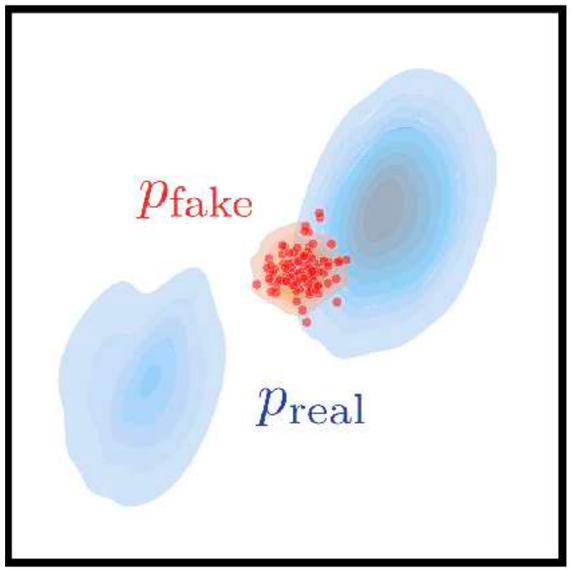
Reverse KL problem

How to avoid mode collapse?



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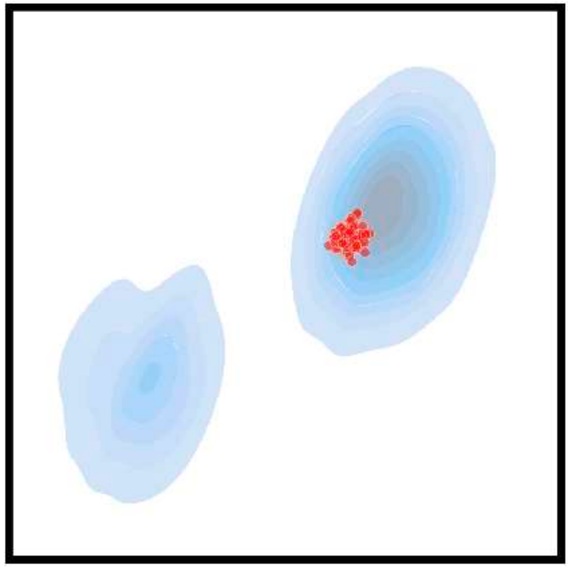
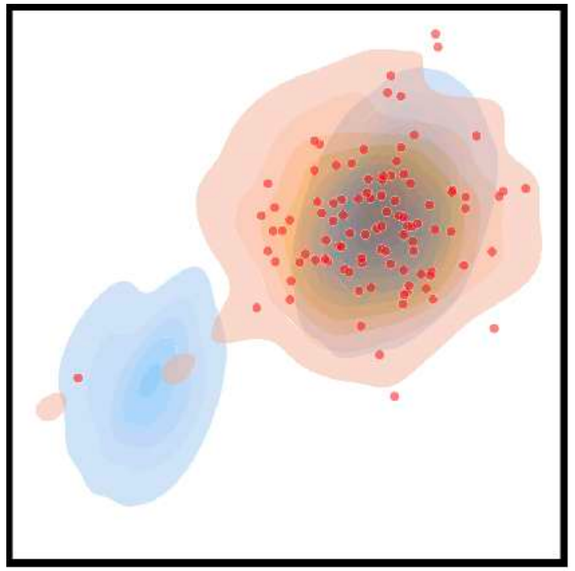
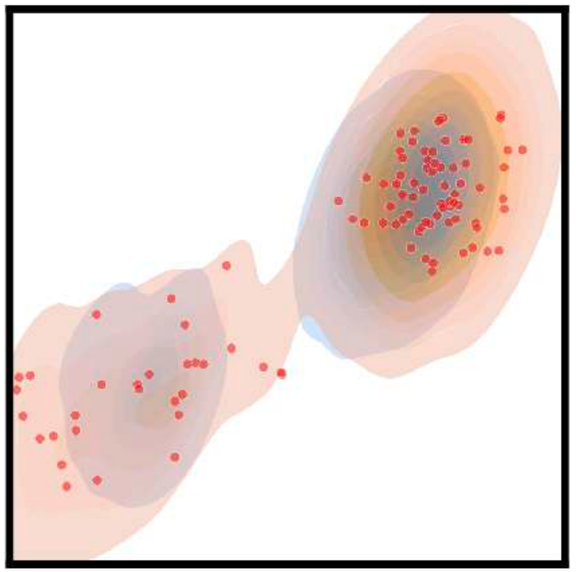


Real score only



Real + fake scores



?

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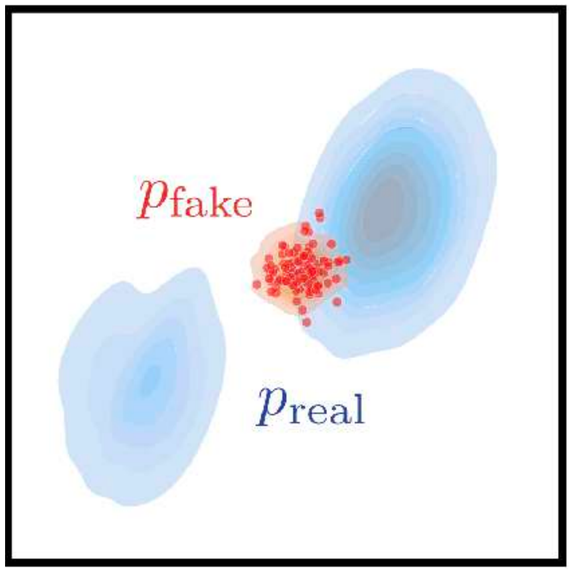
Reverse KL problem

Additionally use a classical Knowledge Distillation approach



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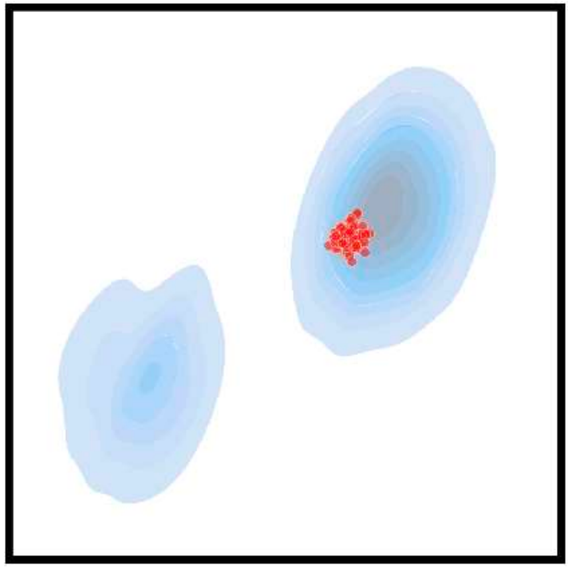

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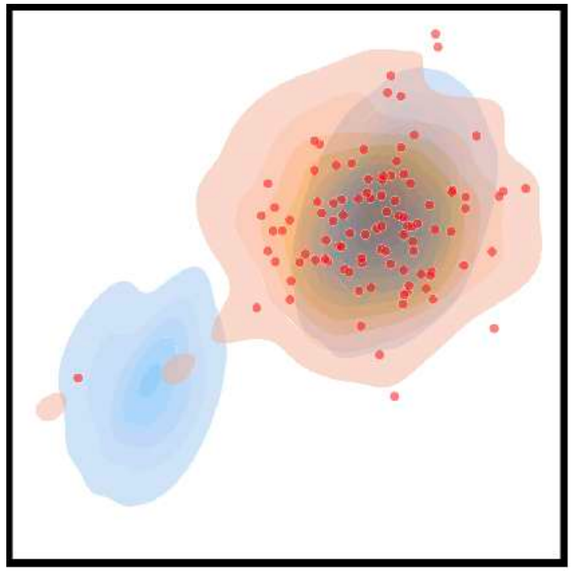
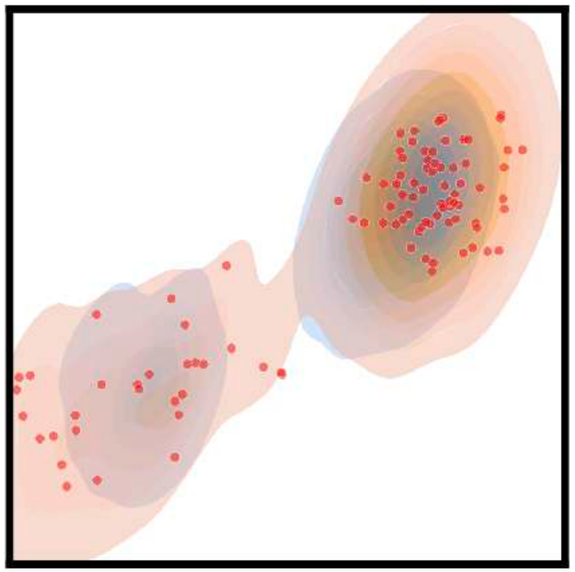


Real + fake scores



Real + fake scores + KD

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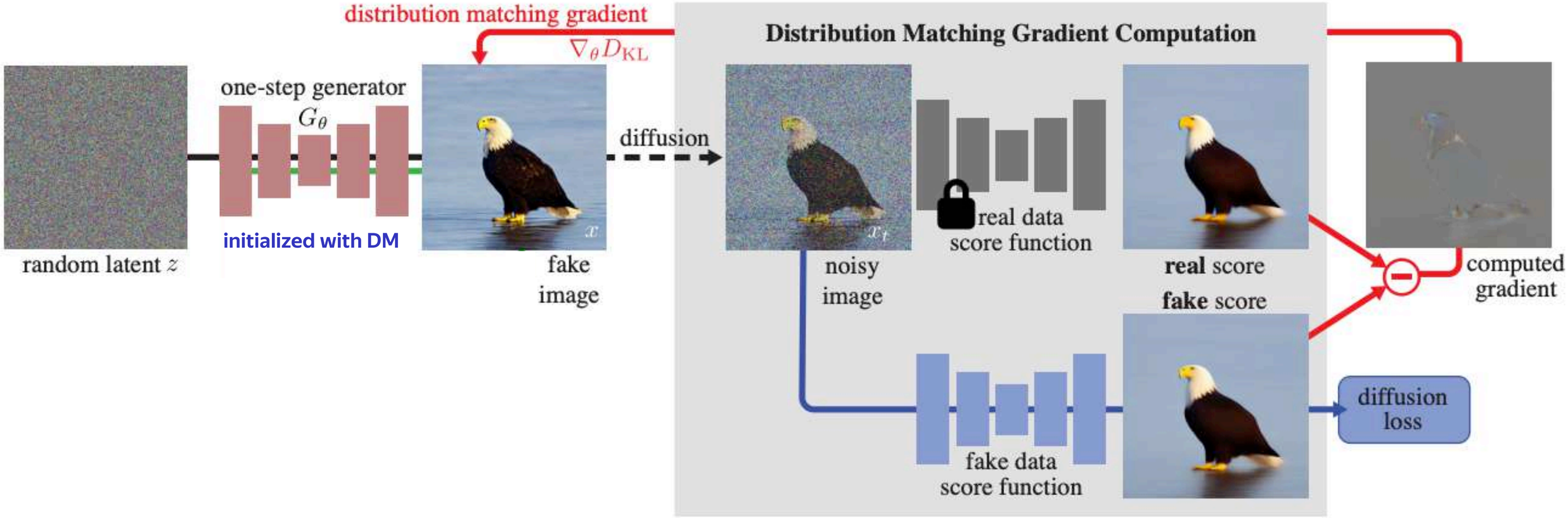
Distribution Matching Distillation



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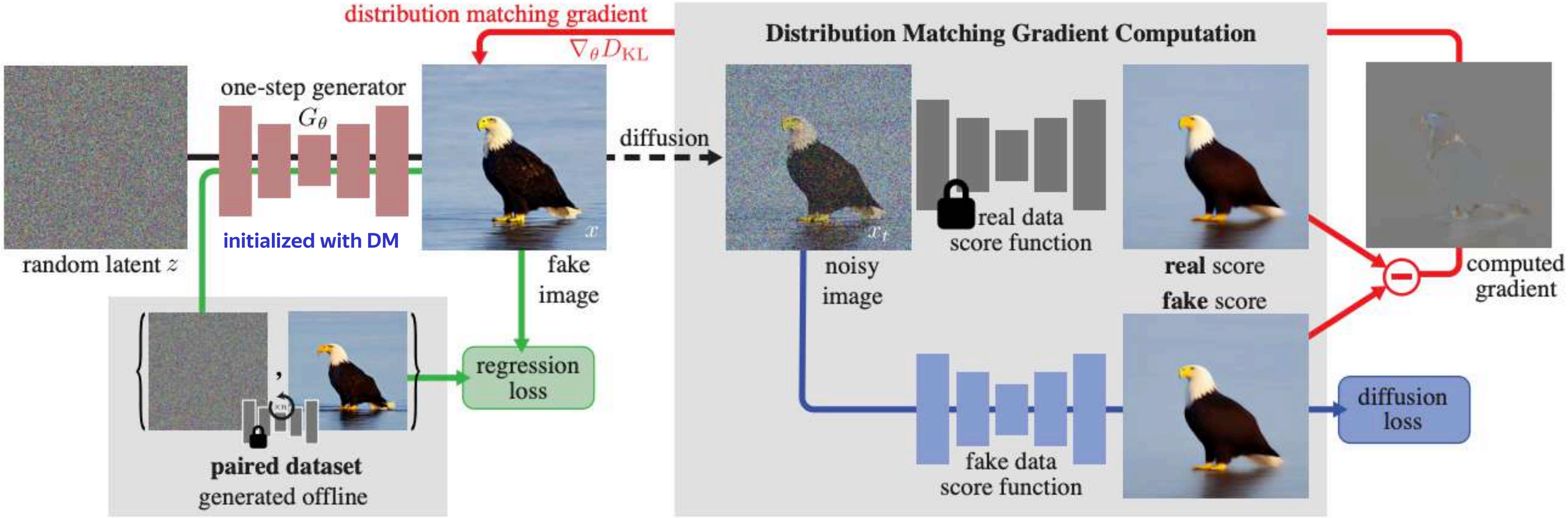

T. Yin, et al, One-step Diffusion with Distribution Matching Distillation. 2024

Score Distillation Sampling | SDS

Can we get rid of the fake DM?

B. Poole, et al., DreamFusion: Text-to-3D using 2D Diffusion. 2023

Score Distillation Sampling | SDS

Can we get rid of the fake DM?

Let's use unbiased score function estimation for $s_{fake}(\mathbf{x}_t, t)$ on $\mathbf{x} \sim G_\theta(\mathbf{z})$:

$$s_{fake}(\mathbf{x}_t, t) = \nabla_{\mathbf{x}_t} \log p_{fake}(\mathbf{x}_t) = \mathop{\mathbb{E}}_{\mathbf{x} \sim p_{fake}} [\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t \,|\, \mathbf{x}) \,|\, \mathbf{x}_t] \approx \nabla_{x_t} \log q(\mathbf{x}_t \,|\, \mathbf{x}) = -\frac{\mathbf{x}_t - \alpha_t \mathbf{x}}{\sigma_t^2}$$

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— Performs poorly alone → used only as a minor polishing loss

— Alternative cheap ways to avoid the fake DM are still unsuccessful

B. Poole, et al., DreamFusion: Text-to-3D using 2D Diffusion. 2023

DMD summary

Key idea: use DM forward passes to estimate $\nabla_{\mathbf{x}} \log p_{fake}(\mathbf{x})$ and $\nabla_{\mathbf{x}} \log p_{real}(\mathbf{x})$

Pros

— Fruitful idea for future research and promising results

Cons

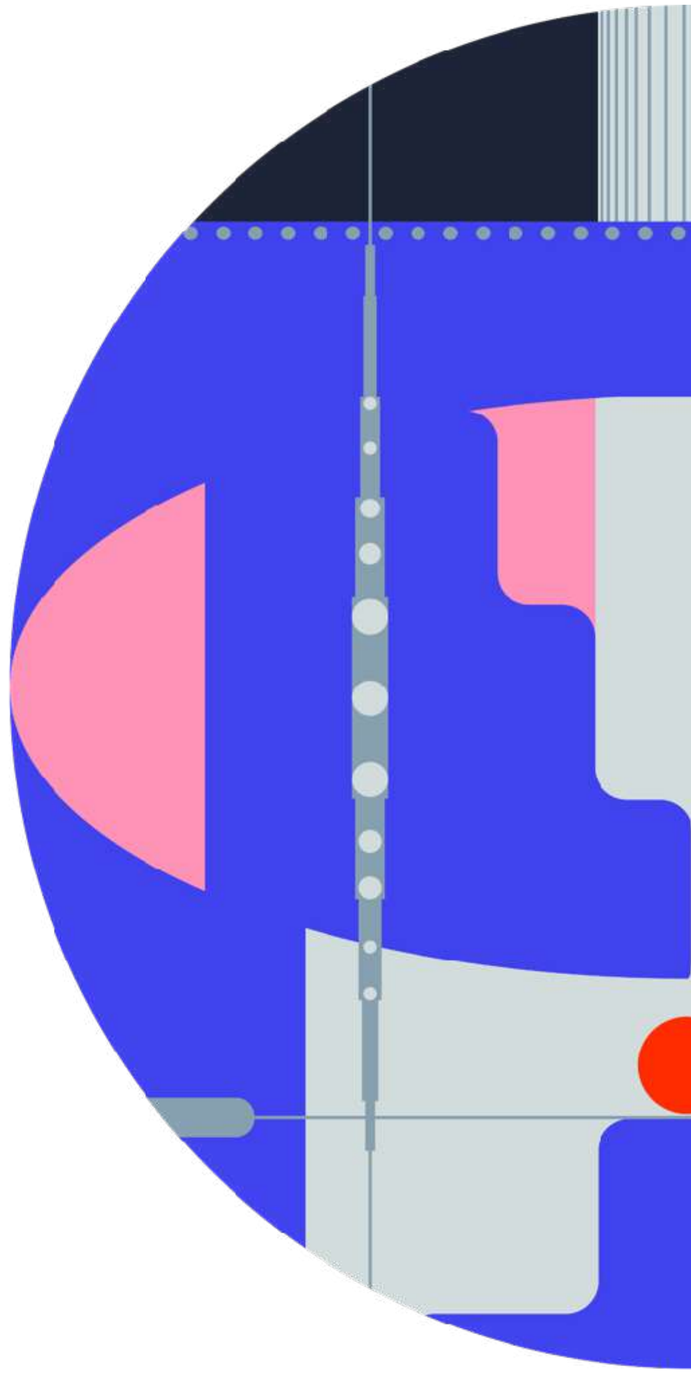
— Requires additional DM training in parallel

— Relatively expensive and unstable

— Prone to mode collapse → needs additional losses

T. Yin, et al, Improved Distribution Matching Distillation for Fast Image Synthesis. 2024

Adversarial Diffusion Distillation

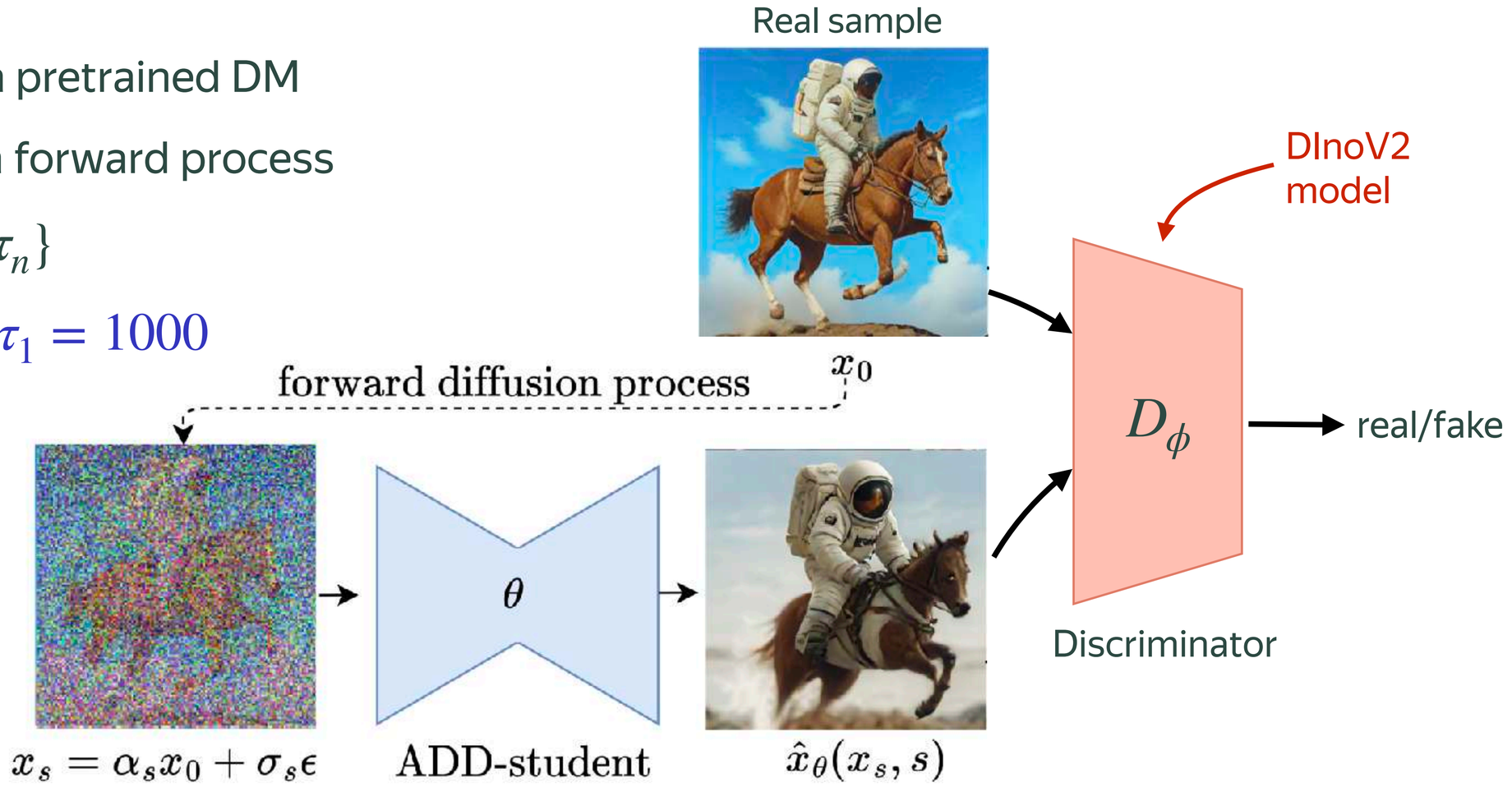


Adversarial Diffusion Distillation

1. $G_\theta(\mathbf{z})$ is initialized with a pretrained DM
2. Multistep sampling via a forward process

$$s \in T_{student} = \{\tau_1, \ldots, \tau_n\}$$

ADD = GAN for $n = 1$ and $\tau_1 = 1000$



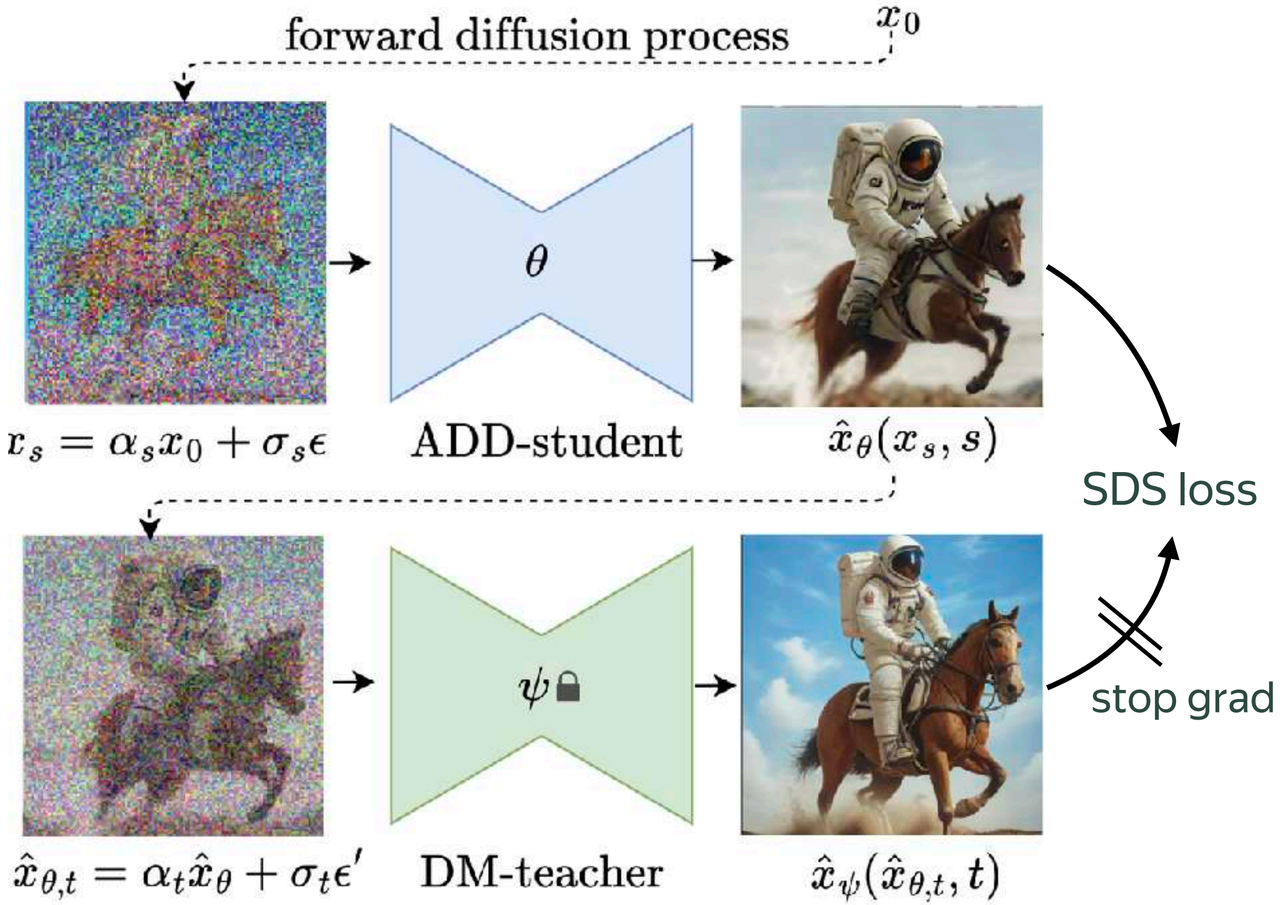
A. Sauer et al., Adversarial Diffusion Distillation. 2023

Adversarial Diffusion Distillation

Combine GAN and SDS losses

Here SDS is reparameterized via $\mathbf{x}_0$-prediction

SDS plays a minor role in the final performance

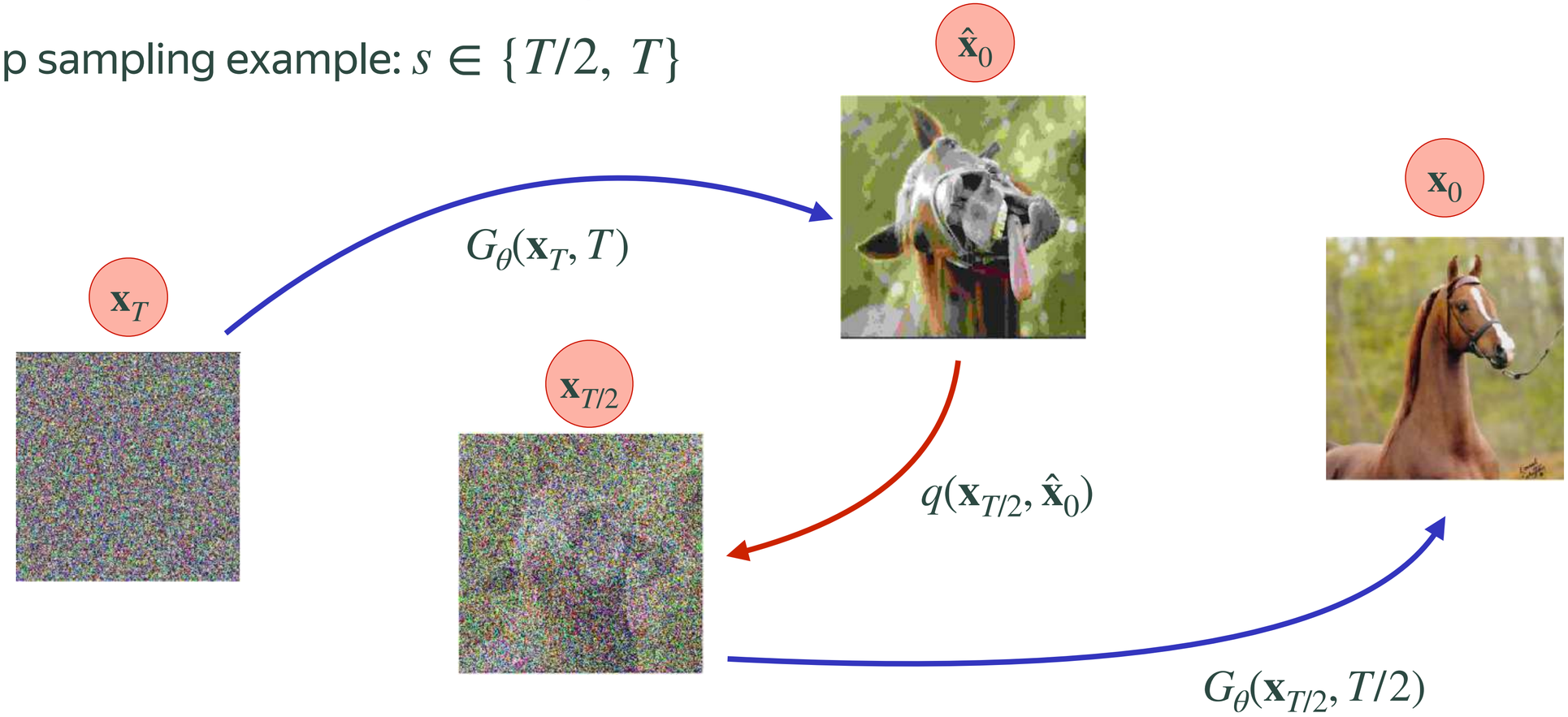


A. Sauer et al., Adversarial Diffusion Distillation. 2023

ADD sampling

Identical to sampling in Consistency Models

2-step sampling example: $s \in \{T/2,\ T\}$



ADD summary

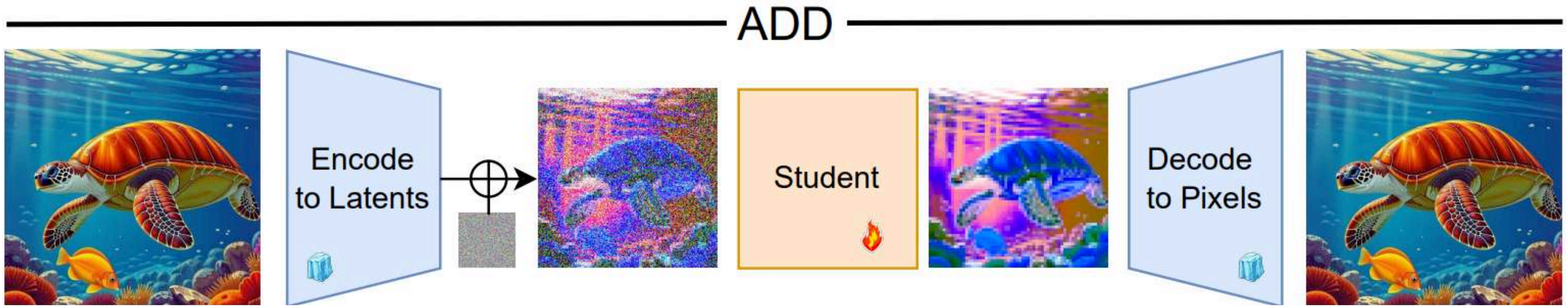
SDXL-Turbo

Key idea: DM-initialized GAN + SDS for negligible polishing

Both objectives are prone to mode collapse → poor sample diversity for a given prompt

A. Sauer et al., Adversarial Diffusion Distillation. 2023

Adversarial Diffusion Distillation



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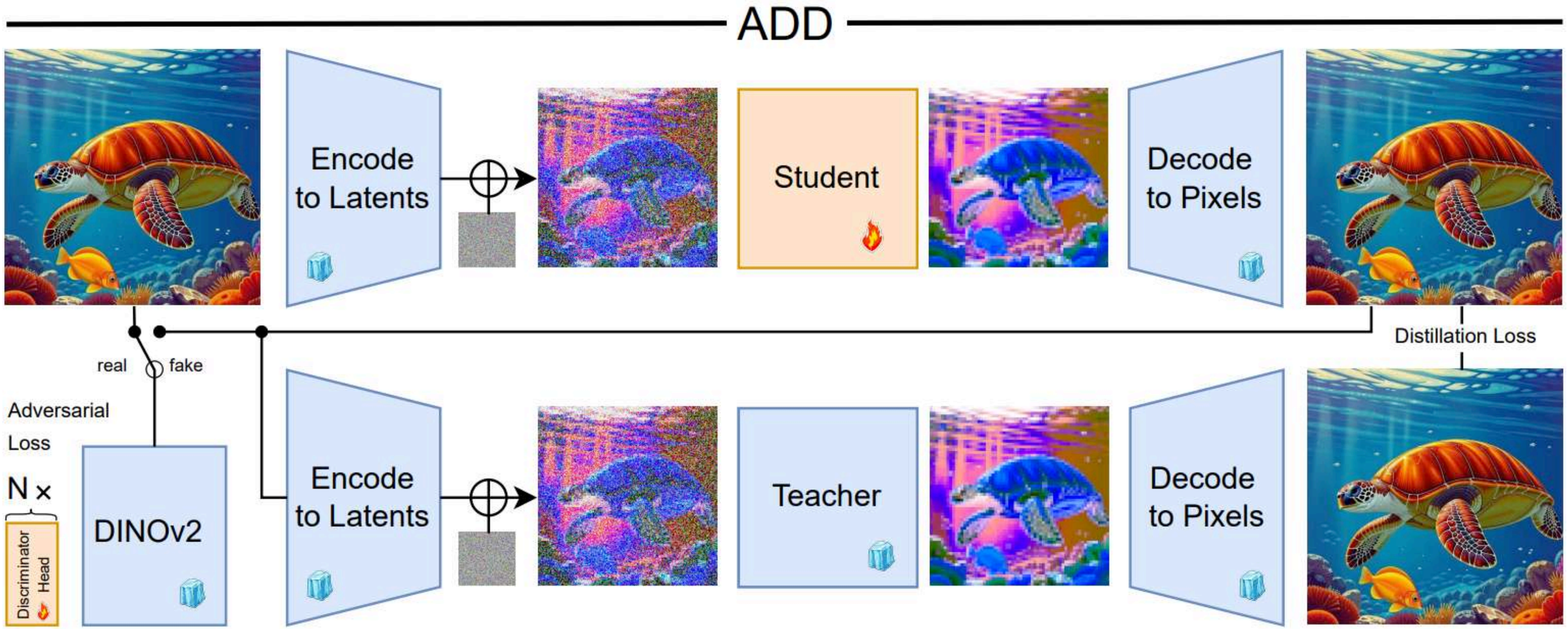
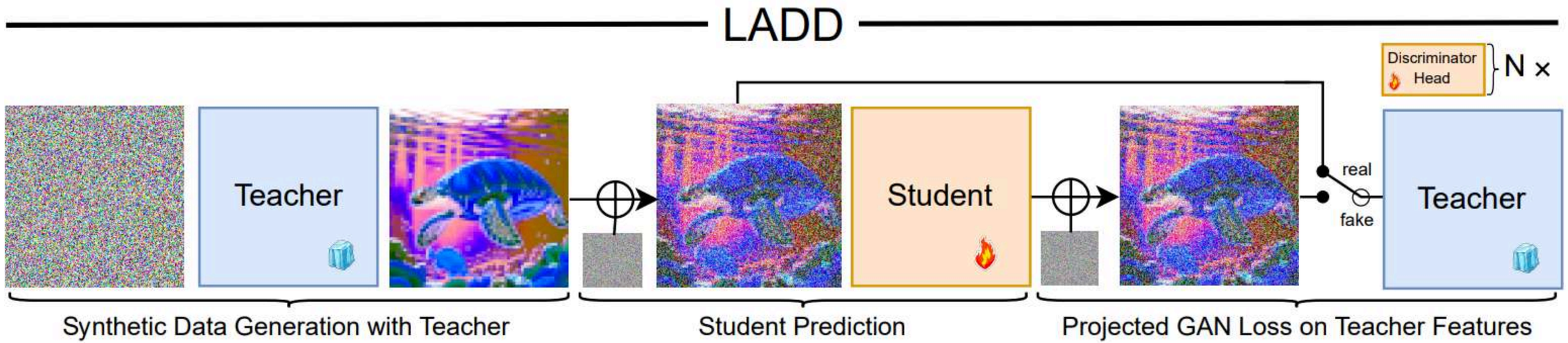
Adversarial Diffusion Distillation



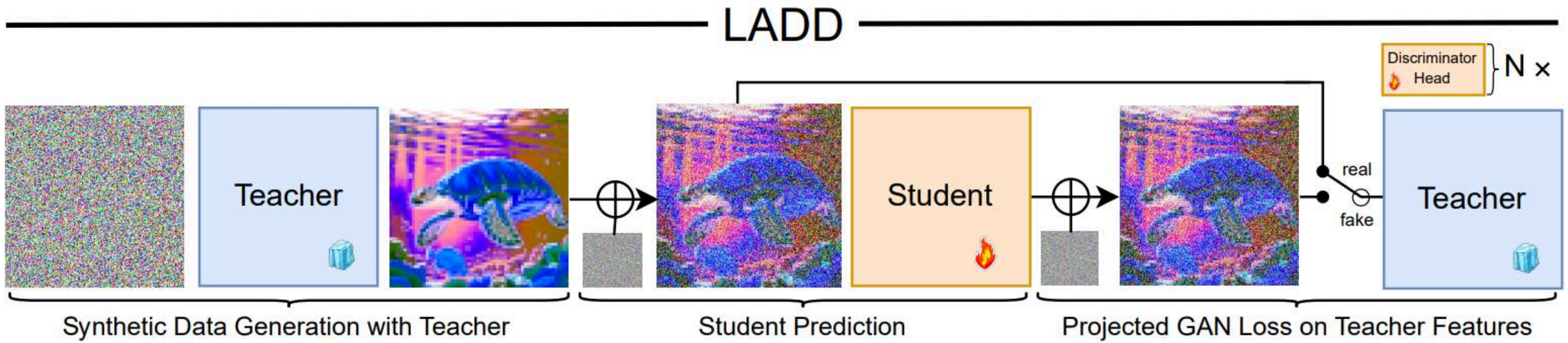
A. Sauer et al., Adversarial Diffusion Distillation. 2023

Latent Adversarial Diffusion Distillation



A. Sauer et al., Fast High-Resolution Image Synthesis with Latent Adversarial Diffusion Distillation. 2024

Latent Adversarial Diffusion Distillation



Practical note: If you have higher quality external data than teacher samples, then use it

A. Sauer et al., Fast High-Resolution Image Synthesis with Latent Adversarial Diffusion Distillation. 2024

LADD vs ADD

Stays in VAE latent space → more efficient training

Discriminator is initialized with a teacher DM → feedback from different noise levels

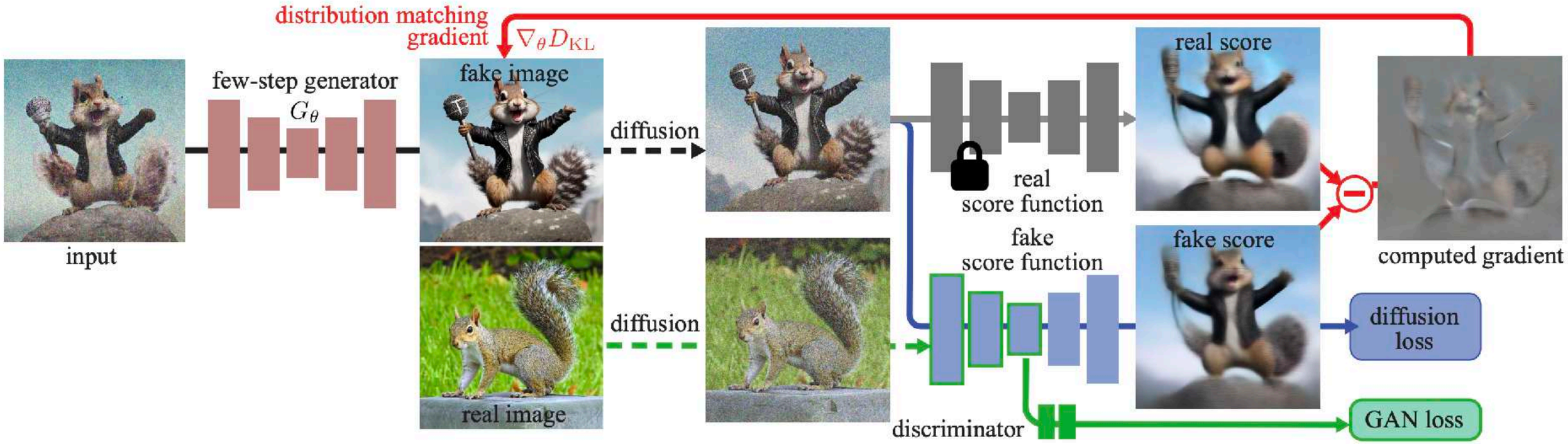
SDS is not needed

LADD is a reasonable choice in production

What about …

Improved DMD | DMD2

Let’s combine GAN and DMD losses



T. Yin, et al, Improved Distribution Matching Distillation for Fast Image Synthesis. 2024

Improved DMD | DMD2

DMD

— KD for stability and mode coverage

— Single step generator

— 1 update step for fake DM

DMD2

— GAN

— Multistep generator

— 5 update steps for fake DM (important)

DMD2 results

4-step DMD2-Qwen-Image

4-step DMD2-SDXL

Takeaways

DMD — Use DMs for reverse KL gradient estimation

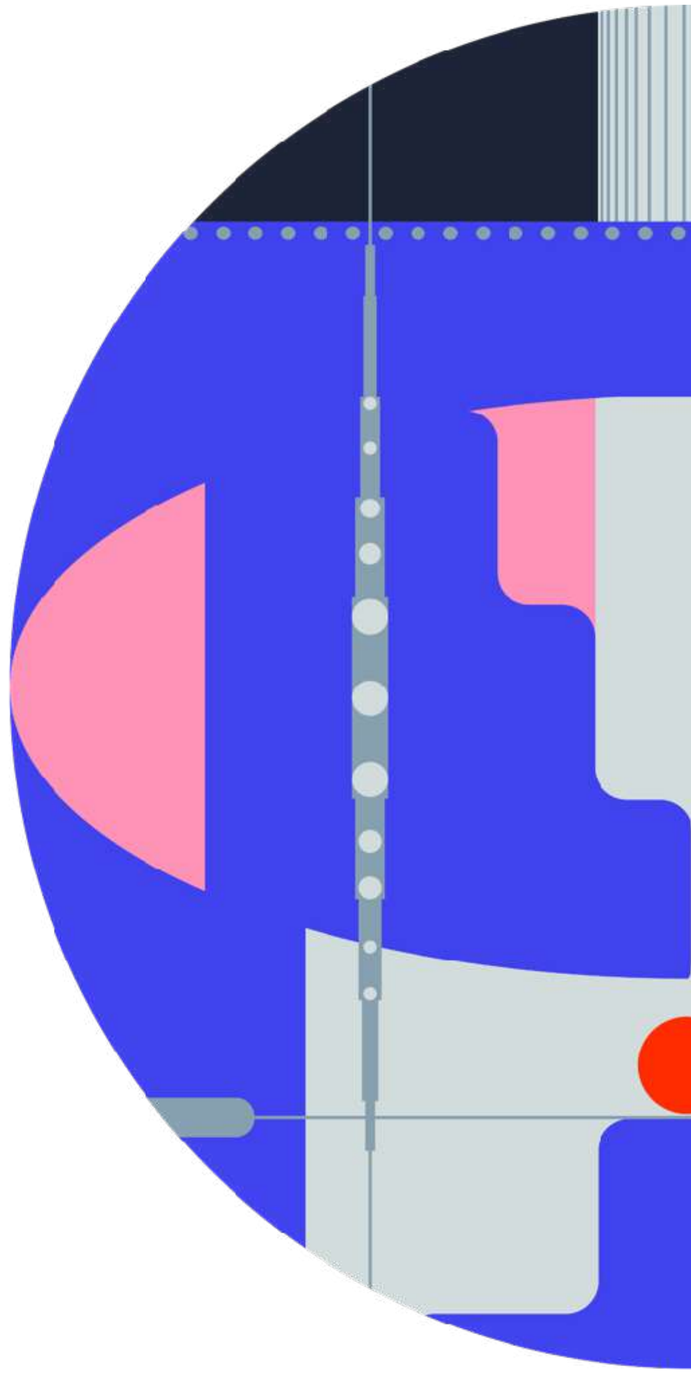
ADD, LADD — Diffusion-based GANs, where DM initialization plays a crucial role

DMD2 — DMD + GAN loss — **current state-of-the-art**

All of them trade diversity for image fidelity but that is what we are okay to trade in practice

All of them can outperform their teachers due to lower coverage

Maximum Mean Discrepancy Distillation



Motivation

Goal: effective and scalable few-step method

w/o *adversarial objectives* and *extra trainable models*

Better and cheaper than flow-map models

Maximum Mean Discrepancy

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}_{\mathbf{x},\mathbf{x}'\sim P}[k(\mathbf{x}, \mathbf{x}')] + \mathbb{E}_{\mathbf{y},\mathbf{y}'\sim Q}[k(\mathbf{y}, \mathbf{y}')] - 2\,\mathbb{E}_{\mathbf{x}\sim P,\,\mathbf{y}\sim Q}[k(\mathbf{x}, \mathbf{y})]$$

$k(\cdot, \cdot)$ — kernel function

In practice, simply a linear combination of all possible pairwise sample distances

Linear $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y}$

Gaussian (RBF) $k(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\sigma^2}\right)$

Energy $k(\mathbf{x}, \mathbf{y}) = -\|\mathbf{x} - \mathbf{y}\| \Rightarrow \mathrm{MMD}^2 = \mathrm{ED}$ — Energy Distance

Y. Li et al., Generative Moment Matching Networks. 2015

Problem with MMD-based models

Problem

Images are high dimensional

curse of dimensionality

Requires too many samples for accurate estimation

Problem with MMD-based models

Problem

Images are high dimensional

↓

curse of dimensionality

↓

Requires too many samples for accurate estimation

Solution

Work in lower dimensional **feature space**

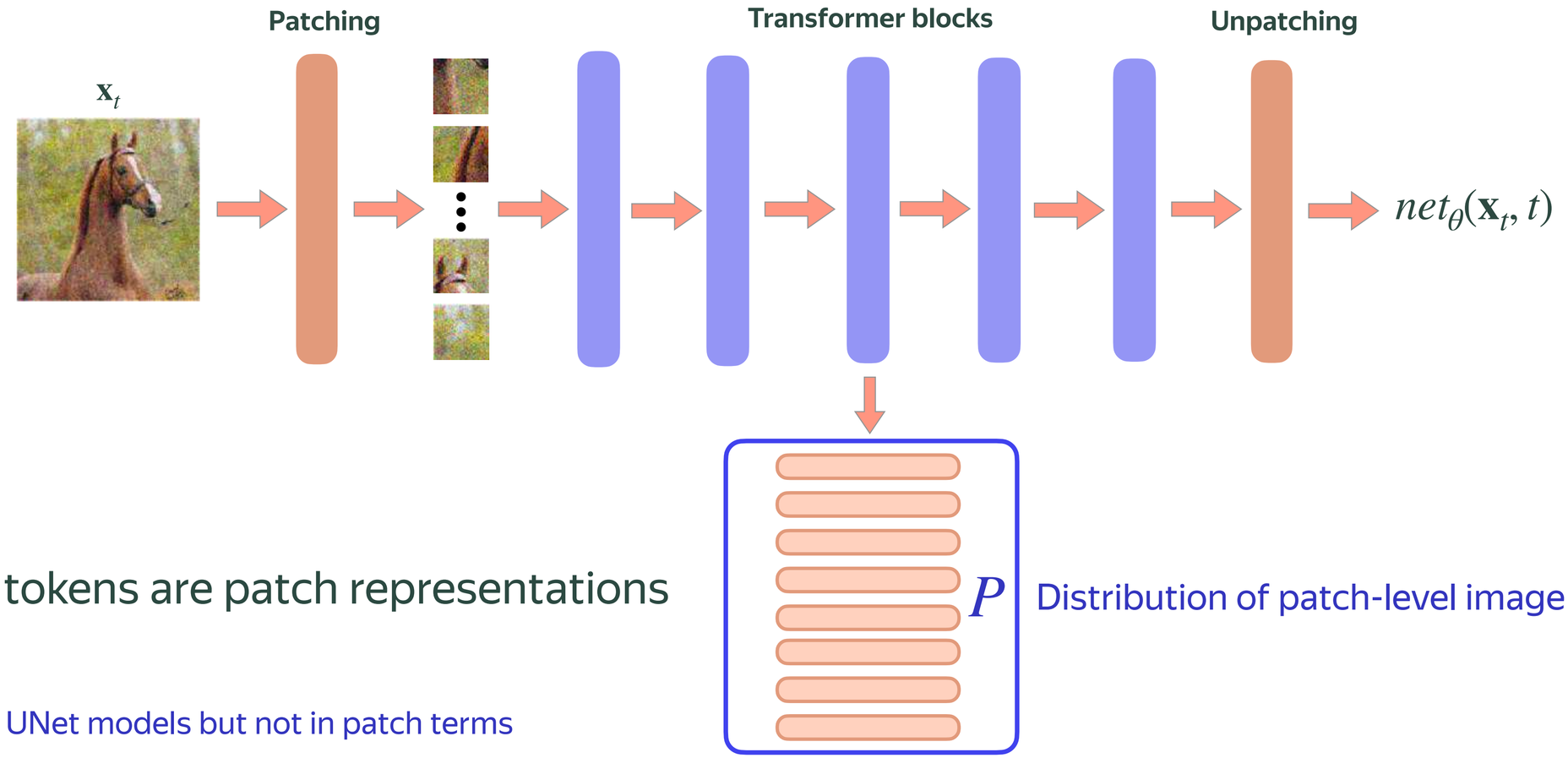
Image distribution → **patch-level** distribution

↓

Much more lower dimension samples for free

For example: 1 image → 256 patches

Diffusion models for feature extraction

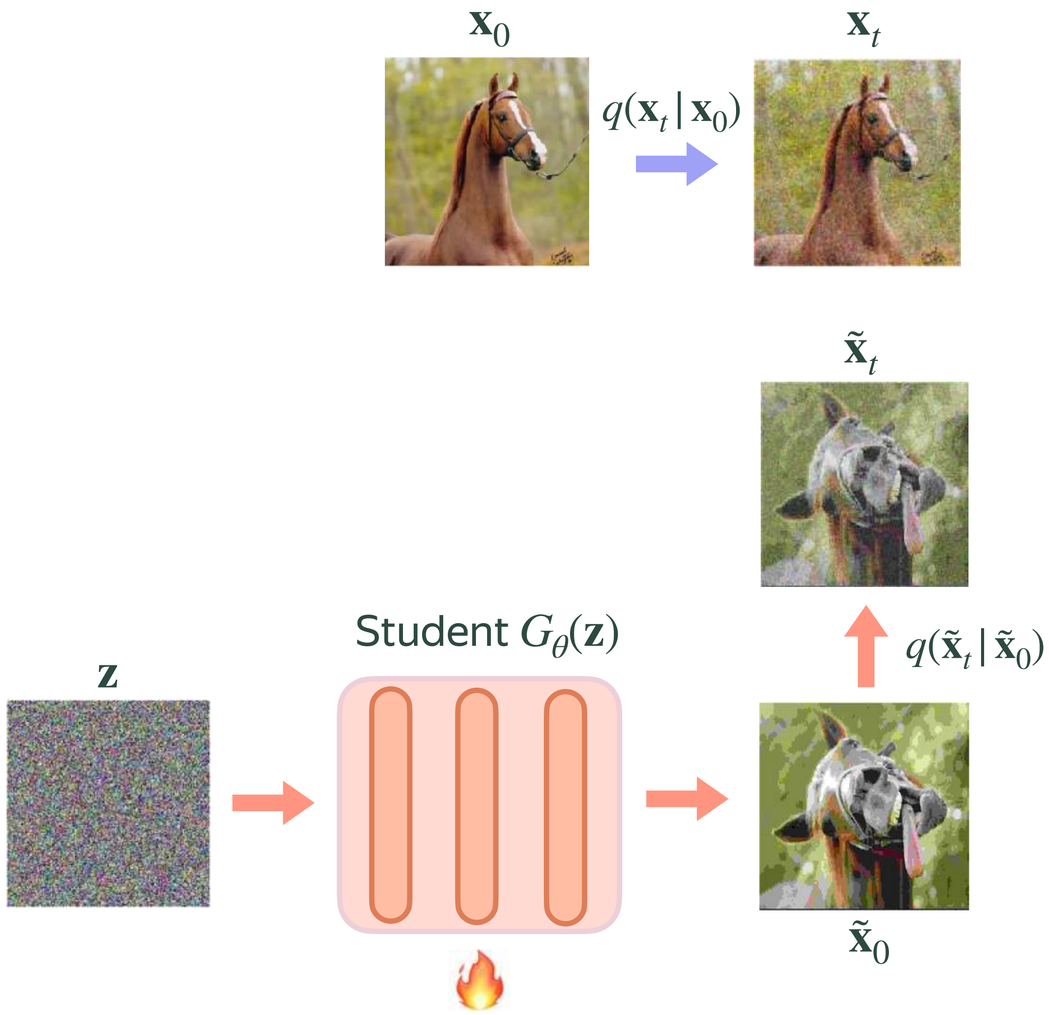


DiT feature tokens are patch representations

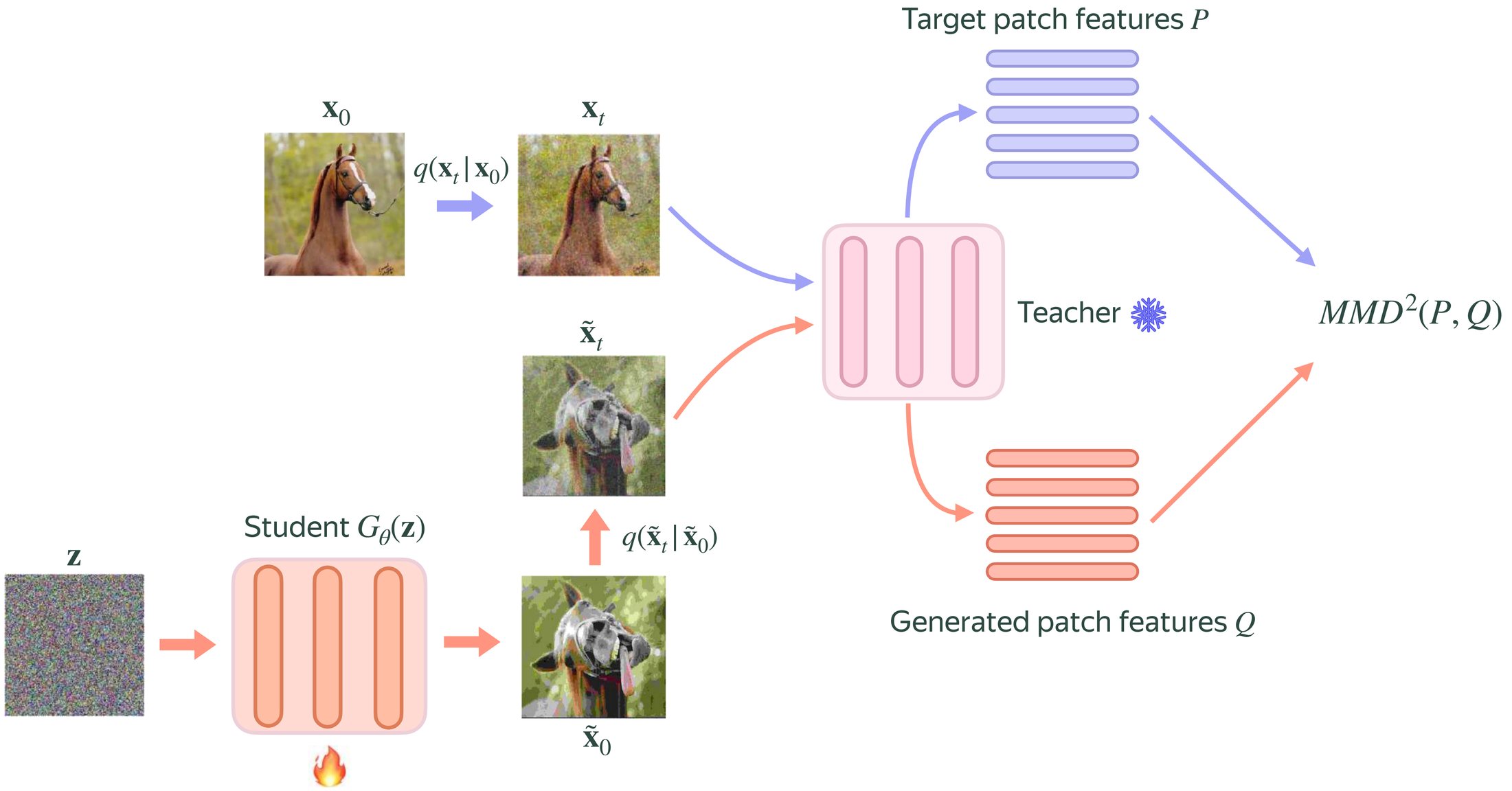
Distribution of patch-level image representations

P.S. similar for the UNet models but not in patch terms

MMD Distillation



MMD Distillation



MMD is a strong standalone objective

DMD2 + MMD loss

MMD loss only

Only slight degradation in defects compared to DMD2

Doesn’t require discriminators or fake DMs

Stable & 7x faster training

Future work: make MMD a self-contained loss

Drifting Models

Drifting Models

Let's define **drifting fields** $\mathbf{V}_{p,q}(\cdot)$

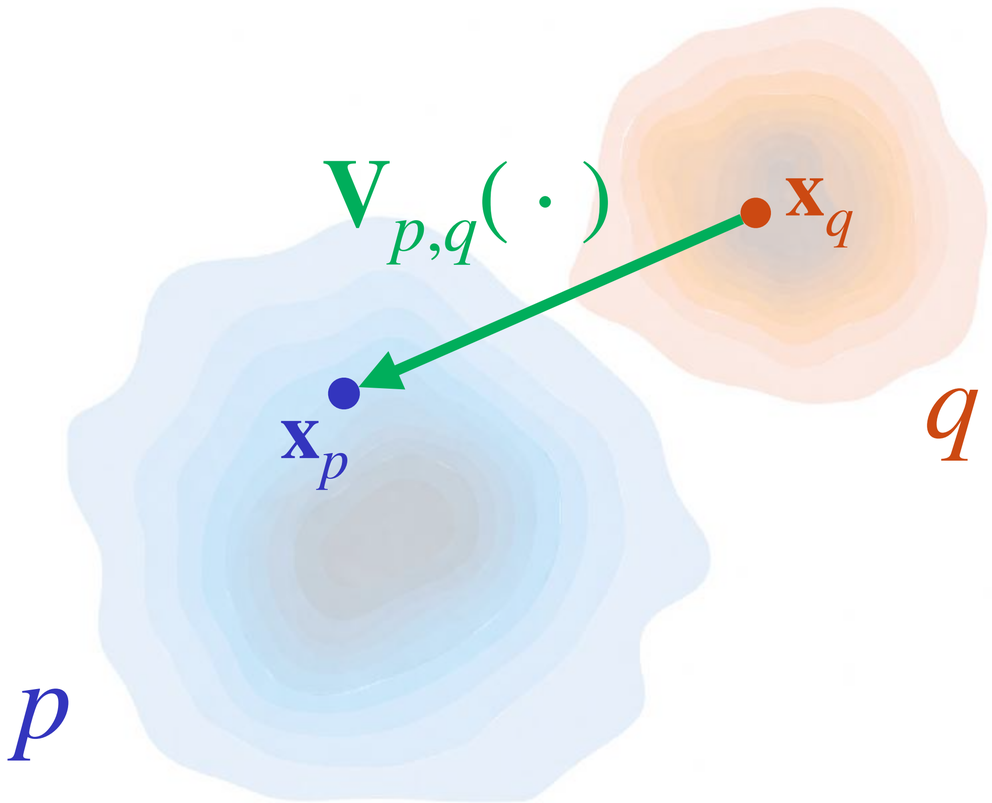
p — data distribution

q — generated distribution

$\mathbf{V}_{p,q}(\cdot)$ — drifting direction from $\mathbf{x}_q \to \mathbf{x}_p$

Equilibrium condition

$$q = p \;\Rightarrow\; \mathbf{V}_{p,q}(\mathbf{x}) = 0,\ \forall \mathbf{x}$$



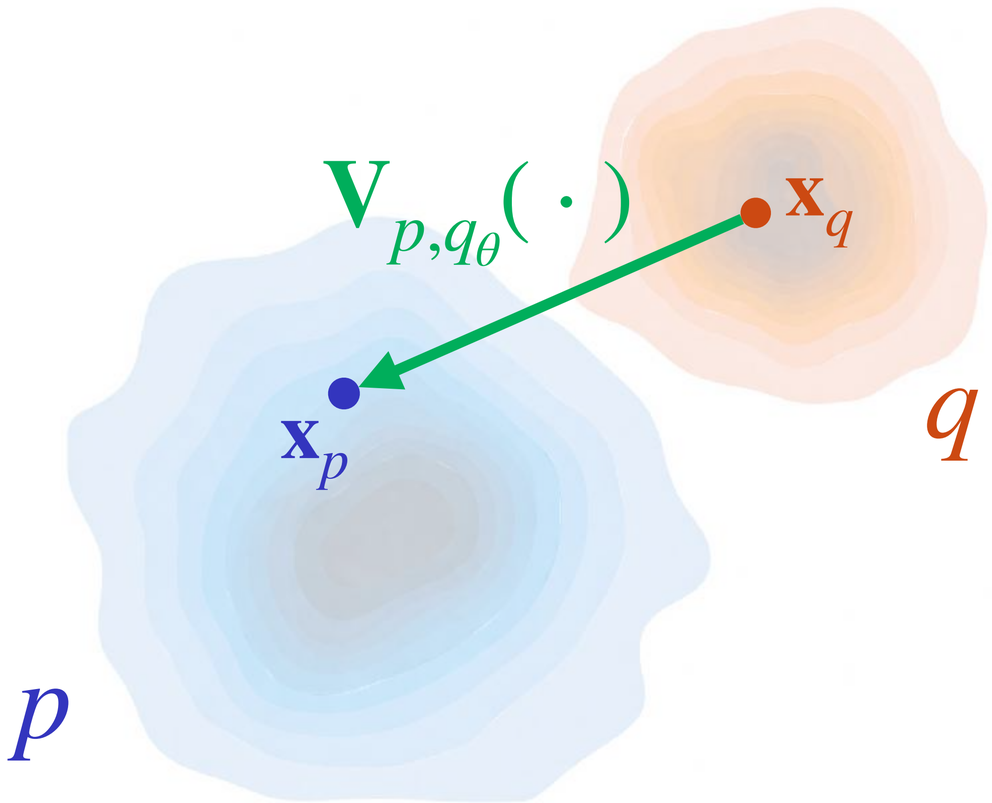
M. Deng et al., Generative Modeling via Drifting. 2026

Drifting Models

Train a generator $G_\theta(\mathbf{z}) \to \mathbf{x}$

$$G_{\theta_{i+1}}(\mathbf{z}) \leftarrow G_{\theta_i}(\mathbf{z}) + \mathbf{V}_{p,q_{\theta_i}}\left(G_{\theta_i}(\mathbf{z})\right)$$

$\mathbf{V}_{p,q_{\theta_i}}\left(G_{\theta_i}(\mathbf{z})\right)$ — “gradient” direction to real distribution



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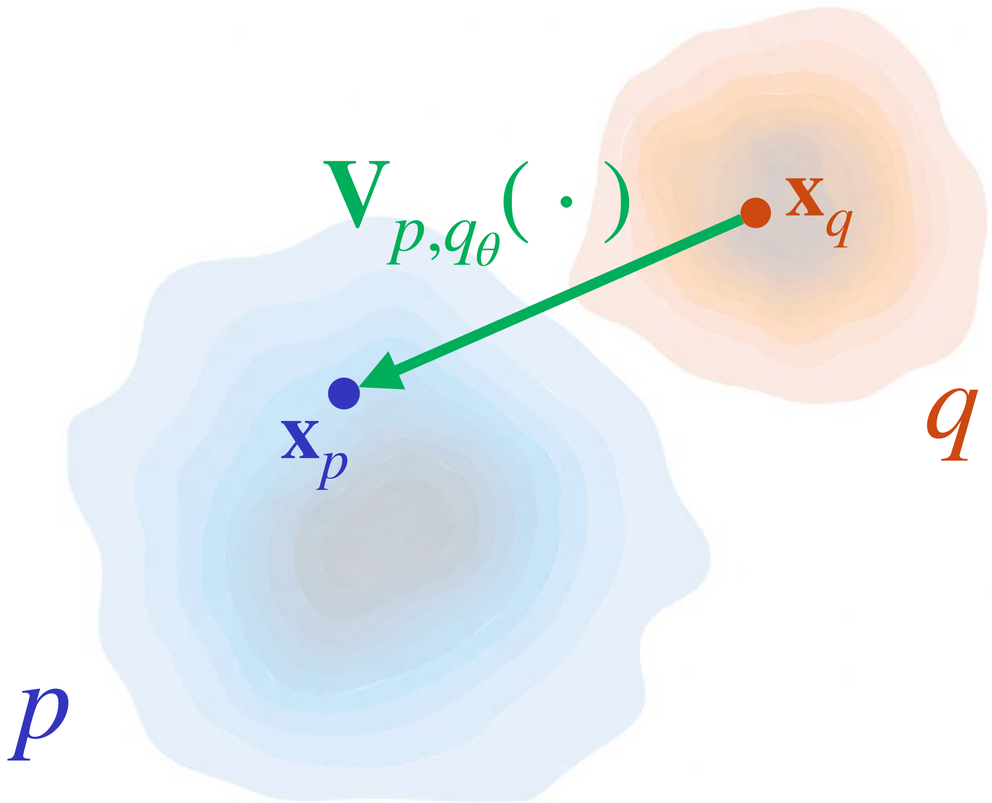
$\mathbf{V}_{p,q_{\theta_i}}\left(G_{\theta_i}(\mathbf{z})\right)$ — "gradient" direction to real distribution

$$L = \mathbb{E}_{\mathbf{z}}\left[\left\| G_\theta(\mathbf{z}) - \mathrm{sg}\left(G_\theta(\mathbf{z}) + \mathbf{V}_{p,q_\theta}\left(G_\theta(\mathbf{z})\right)\right)\right\|^2\right]$$

Hm, looks familiar…



M. Deng et al., Generative Modeling via Drifting. 2026

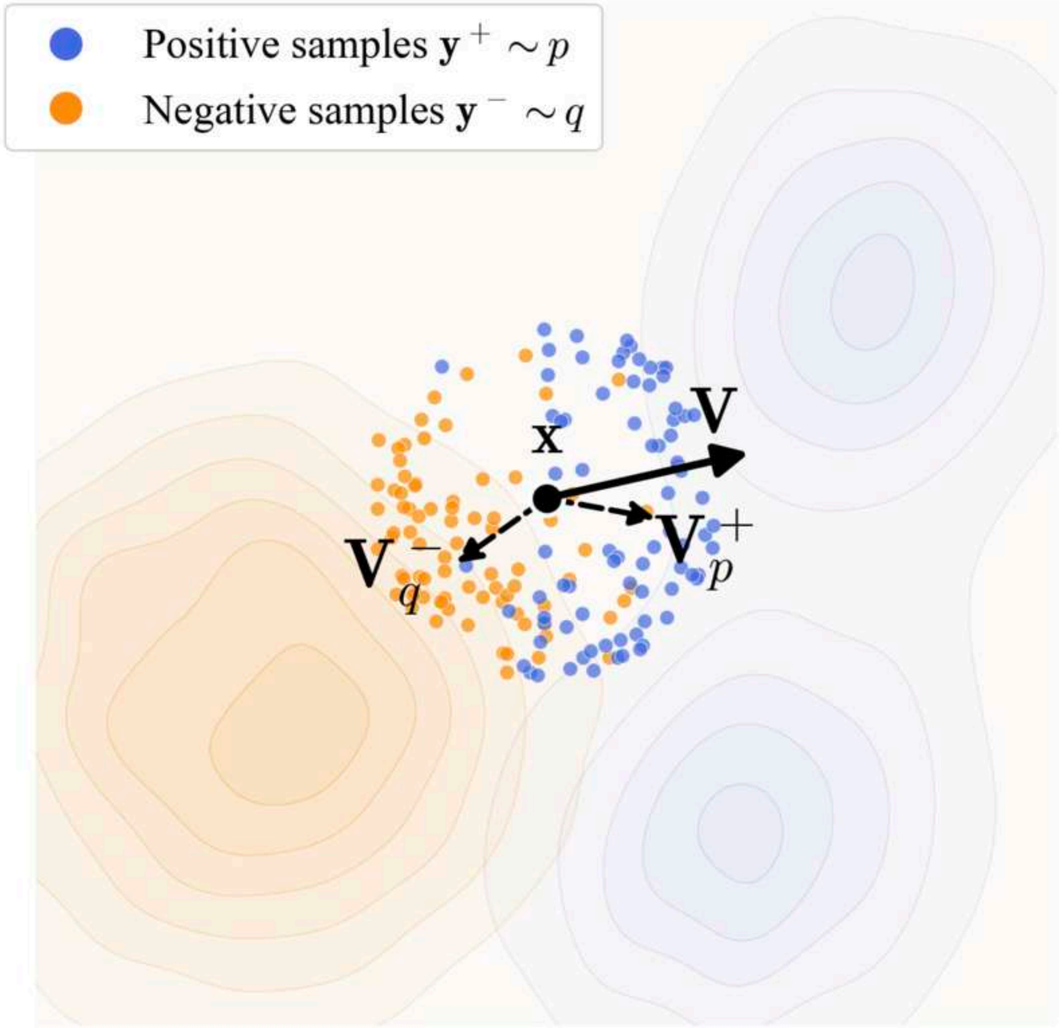
Instantiation of Drifting Models

The proposed variant of the drifting field in the paper

$$\mathbf{V}_{p,q}(\mathbf{x}) = \mathbf{V}^{+}_{p}(\mathbf{x}) - \mathbf{V}^{-}_{q}(\mathbf{x})$$

$\mathbf{V}^{+}_{p}(\mathbf{x})$ — local mean of real samples $\mathbf{y}^{+}$ (positives)

$\mathbf{V}^{-}_{q}(\mathbf{x})$ — local mean of fake samples $\mathbf{y}^{-}$ (negatives)



M. Deng et al., Generative Modeling via Drifting. 2026

Instantiation of Drifting Models

The authors define $\mathbf{V}_p^+(\mathbf{x})$ and $\mathbf{V}_q^-(\mathbf{x})$ as follows:

$$\mathbf{V}_p^+(\mathbf{x}) := \frac{1}{Z_p(\mathbf{x})} \mathbb{E}_p\Big[k\big(\mathbf{x}, \mathbf{y}^+\big)\,\big(\mathbf{y}^+ - \mathbf{x}\big)\Big], \qquad Z_p(\mathbf{x}) := \mathbb{E}_p\Big[k\big(\mathbf{x}, \mathbf{y}^+\big)\Big]$$

$$\mathbf{V}_q^-(\mathbf{x}) := \frac{1}{Z_q(\mathbf{x})} \mathbb{E}_q\Big[k\big(\mathbf{x}, \mathbf{y}^-\big)\,\big(\mathbf{y}^- - \mathbf{x}\big)\Big], \qquad Z_q(\mathbf{x}) := \mathbb{E}_q\Big[k\big(\mathbf{x}, \mathbf{y}^-\big)\Big]$$

$$k(\mathbf{x}, \mathbf{y}) = e^{-\frac{1}{\tau}||\mathbf{x}-\mathbf{y}||^2}$$

Gaussian kernel

M. Deng et al., Generative Modeling via Drifting. 2026

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$$k(\mathbf{x}, \mathbf{y}) = e^{-\frac{1}{\sigma^2}||\mathbf{x}-\mathbf{y}||^2}$$

Gaussian kernel

↓ Some magic algebra

$$V_{p,q}(\mathbf{x}) = \mathbf{V}_p^+(\mathbf{x}) - \mathbf{V}_q^-(\mathbf{x}) = \frac{1}{Z_p Z_q} \mathbb{E}_{p,q}\Big[k(\mathbf{x}, \mathbf{y}^+) k(\mathbf{x}, \mathbf{y}^-)(\mathbf{y}^+ - \mathbf{y}^-)\Big]$$

M. Deng et al., Generative Modeling via Drifting. 2026

Connection to DMD

What if I say that **this instantiation is equivalent to DMD but with ideal denoisers?**

Ideal denoiser: $$D^*(\mathbf{x};\sigma) = \frac{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i, \sigma^2\mathbf{I})\,\mathbf{y}_i}{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i, \sigma^2\mathbf{I})}$$

Connection to DMD

Let's replace $k(\mathbf{x}, \mathbf{y}) = e^{-\frac{1}{\sigma^2}||\mathbf{x}-\mathbf{y}||^2}$ with $\mathcal{N}(\mathbf{x};\ \mathbf{y}, \sigma^2\mathbf{I}) \Rightarrow$

$$k(\mathbf{x}, \mathbf{y}^+) = \mathcal{N}(\mathbf{x};\ \mathbf{y}^+, \sigma^2\mathbf{I}) \qquad Z_p(\mathbf{x}) = \mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)] = \sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^+, \sigma^2\mathbf{I})$$

$$k(\mathbf{x}, \mathbf{y}^-) = \mathcal{N}(\mathbf{x};\ \mathbf{y}^-, \sigma^2\mathbf{I}) \qquad Z_q(\mathbf{x}) = \mathbb{E}_q[k(\mathbf{x}, \mathbf{y}^-)] = \sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})$$

Connection to DMD

$$k(\mathbf{x}, \mathbf{y}^+) = \mathcal{N}(\mathbf{x};\ \mathbf{y}^+, \sigma^2\mathbf{I}) \qquad Z_p(\mathbf{x}) = \mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)] = \sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^+, \sigma^2\mathbf{I})$$

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$$D^*_{real}(\mathbf{x};\sigma) = \frac{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^+, \sigma^2\mathbf{I})\,\mathbf{y}_i^+}{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^+, \sigma^2\mathbf{I})}$$

$$D^*_{fake}(\mathbf{x};\sigma) = \frac{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})\,\mathbf{y}_i^-}{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})}$$

Connection to DMD

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$$D^*_{fake}(\mathbf{x};\sigma) = \frac{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})\,\mathbf{y}_i^-}{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})} = \frac{\sum_i k(\mathbf{x}, \mathbf{y}^-)\mathbf{y}^-}{Z_q(\mathbf{x})}$$

Connection to DMD

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$$D^*_{fake}(\mathbf{x};\sigma) = \frac{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})\,\mathbf{y}_i^-}{\sum_i \mathcal{N}(\mathbf{x};\ \mathbf{y}_i^-, \sigma^2\mathbf{I})} = \frac{\sum_i k(\mathbf{x}, \mathbf{y}^-)\mathbf{y}^-}{Z_q(\mathbf{x})} = \frac{\mathbb{E}_q\left[k(\mathbf{x}, \mathbf{y}^-)\mathbf{y}^-\right]}{Z_q(\mathbf{x})}$$

Objective manipulations

$$V_{p,q}(\mathbf{x}) = \frac{1}{Z_p Z_q} \mathbb{E}_{p,q}\left[k(\mathbf{x}, \mathbf{y}^+)k(\mathbf{x}, \mathbf{y}^-)(\mathbf{y}^+ - \mathbf{y}^-)\right] =$$

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$$= \frac{D^*_{real}(\mathbf{x},\sigma)}{Z_q}\underbrace{\mathbb{E}_q\left[k(\mathbf{x},\mathbf{y}^-)\right]}_{Z_q} - \frac{D^*_{fake}(\mathbf{x},\sigma)}{Z_p}\underbrace{\mathbb{E}_p\left[k(\mathbf{x},\mathbf{y}^+)\right]}_{Z_p} =$$

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$$= \frac{D^*_{real}(\mathbf{x},\sigma)}{Z_q}\underbrace{\mathbb{E}_q\left[k(\mathbf{x},\mathbf{y}^-)\right]}_{Z_q} - \frac{D^*_{fake}(\mathbf{x},\sigma)}{Z_p}\underbrace{\mathbb{E}_p\left[k(\mathbf{x},\mathbf{y}^+)\right]}_{Z_p} = \boxed{D^*_{real}(\mathbf{x},\sigma) - D^*_{fake}(\mathbf{x},\sigma)}$$

DMD with D^* for a fixed small σ

Summary

Particular instantiation in the paper

The proposed drifting models is a **non-parametric DMD** (w/o pretrained real and fake diffusion models)

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2) Requires large batch sizes to be effective even on ImageNet

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The proposed drifting models is a **non-parametric DMD** (w/o pretrained real and fake diffusion models)

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1) In the feature space of some external encoder (e.g., DINOv3)

2) Requires large batch sizes to be effective even on ImageNet

Drifting models are similar to the MMD distillation but

1) The loss is computed on image representations instead of the patch ones

2) MMD distillation uses a teacher DM as a feature extractor

Summary

Drifting Models in general

Very general and productive perspective on **all** implicit generative models: GANs, DMD, MMD, etc